UNIVERSITY OF CALIFORNIA

Los Angeles

Factors that contribute to student applicants achieving and maintaining active status through the first academic year: a case study

A thesis submitted in partial satisfaction of the requirements for the degree of Master of Applied Statistics

by

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ABSTRACT OF THE THESIS

Factors that contribute to student applicants achieving and maintaining active status through the first academic year: a case study

by

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Master of Applied Statistics

University of California, Los Angeles, 2023

Professor Rick Schoenberg, Chair

Student customer contacts with a small private arts college were tracked via touchpoints with the marketing arm of the college. These initial contacts would progress (or not) to status of applicants, enrollees, active students, and active students completing their first year. The objective of this study was to estimate the likelihood of progression from application to enrollment to active status and completion of the first academic year. These likelihoods were estimated by exploratory data analysis using demographics as well as logistic regression methods. The results of this study will be used to direct the marketing department and student advising department toward optimal allocation of resources to improve customer retention. On the marketing side, the study identified poorly performing resource intensive touchpoints as ranked against better performing (and ripe for expansion) cheaper touchpoints. For the student

advisement department, factors related to increased odds of customers dropping off the pathway at each step were identified. These factors varied from step-to-step. Traditionally under-represented minority groups had the greatest odds of dropping off early in the pathway and male students were at greatest risk in the later steps.

The thesis of Christopher Thomas Bowne is approved.

Hongquan Xu

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University of California, Los Angeles

2023

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CHAPTER 1. INTRODUCTION, BACKGROUND AND QUESTIONS TO BE ANSWERED

This study takes place in the context of a small private for-profit art and design college and was motivated by observations from both the marketing department and the student advisement department regarding declining enrollment in recent years. The school had only just put into place a department and staff capable of data analytics and new data tracking systems that allowed for sophisticated data analysis. A pilot study was proposed to better understand the Status Pathway for Students as illustrated in Figure 1 below.

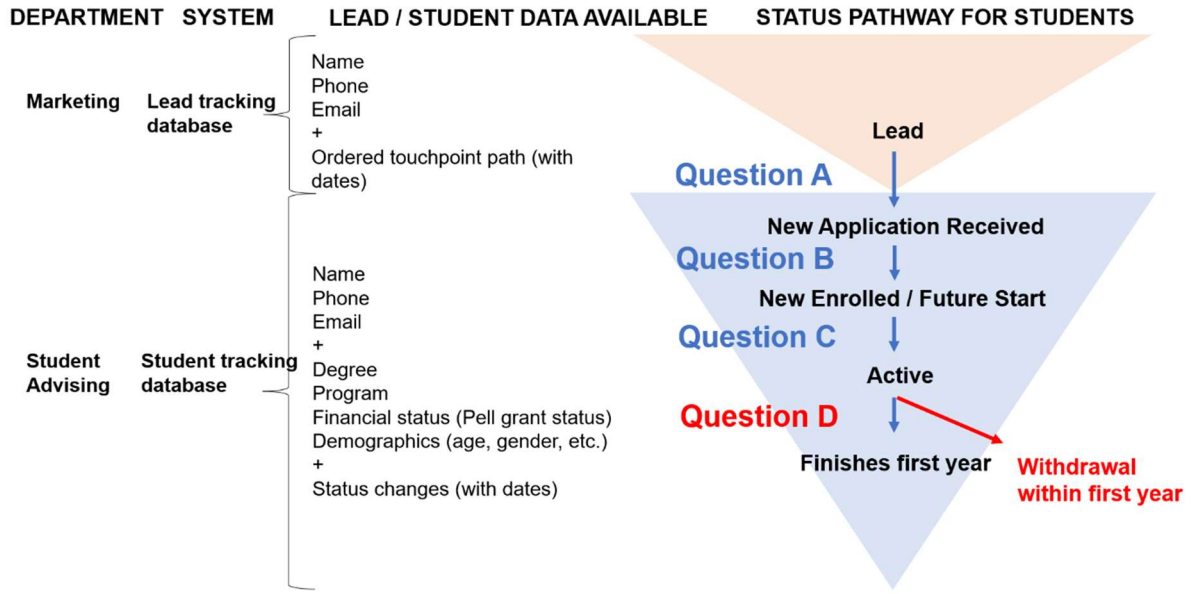

Figure 1. Status Pathway for Students

The status pathway is split between two departments: marketing and student advising. All students start as leads that interface via touchpoints with marketing's lead tracking database. Leads are identified by name, phone, and/or email address and an ordered list (by date) of touchpoints (called the touchpoint path or trail) is stored for each lead. Some of these leads will apply to the school by submitting an application and paying a fee. Once the application is submitted a record is created in the student tracking database that includes their contact information like email along with demographic and financial information (see Table 3) that was

included in the application. The applicant is assigned the status of New Application Received. If the applicant accepts admission and enrolls in courses, they become an enrollee and their status updates to New Enrolled / Future Start. Then if they are recorded as present in their first enrolled course during the first week of the quarter, then their status is updated to Active. Finally, they either finish their first year as students or they withdraw. If they withdraw, their status updates to Withdrawal with a status change timestamp within a year that their status initially updated to Active. Otherwise their status remains Active.

The broad question this study hopes to answer is: what factors are associated with a lead traversing the pathway from lead through New Application Received, New Enrolled / Future Start and Active status, and withdrawing their first year?

This question will be split into four sub-questions* that will be addressed sequentially in each subsequent chapter:

A) What factors are associated with a lead subsequently applying?
B) What factors are associated with an applicant progressing to enrollee?
C) What factors are associated with an enrollee progressing to active student status?
D) What factors are associated with an active student withdrawing within the first year?

* The lettering nomenclature A), B), etc. will be used in reference to these specific sub-questions throughout the study, and from here on they will be called “questions” rather than “sub-questions”. The original question posed will stay the “broad” question.

Question A) What factors are associated with a lead subsequently applying?

This question is of primary concern to the marketing department and this study endeavors to assist them in streamlining by analyzing the performance of different marketing efforts that develop customer contacts so cost-cutting measures can be taken. These customer contacts leave what is called a touchpoint trail, where each point of contact the customer has with the school is identified by a touchpoint.

Question B) What factors are associated with an applicant progressing to enrollee?

Both marketing and student advising are interested in this question. The student advising department wishes to identify demographic and financial factors in the prospective student population that keep them persisting through the application to enrollment status. Marketing would like to know if some of their most costly touchpoints are associated with an applicant progressing to enrollment status (and increasing their likelihood of becoming billable active students).

Question C) What factors are associated with an enrollee progressing to active student status?

Both marketing and student advising are interested in this question. The student advising department wishes to identify demographic and financial factors in the prospective student population that keep them persisting through the enrollment status to active student status. Marketing would like to know if some of their most costly touchpoints are associated with an enrollee becoming a billable active student.

Question D) What factors are associated with an active student withdrawing within the first year?

This question is of interest to student advising for obvious reasons since an income stream ends. By understanding factors associated with withdrawing, student advising may be able to identify students at risk early on and intervene before they withdraw.

By answering all these questions, the school hopes to reallocate resources on the marketing side towards the most productive touchpoints, and on the student advising side, institute flags in the system to mitigate prospective students from dropping out of the pathway towards active status and intervene with active students at risk of dropping out.

CHAPTER 2. DEFINITION OF TERMS

Active status – The status of a prospective student when they start classes and are recorded as in-class on the first day of their first class. The prospective student becomes a student with active status.

Applied/Applicant – See New Application Received status.

Enrolled/Enrollee – See New Enrolled / Future Start status.

Finishes First Year – The status of a student who has not withdrawn from the school in the first academic year (4 quarters). If they withdraw, their status changes to Withdrawal status.

Lag Time – A term used to describe the time it takes on the status pathway from one status to another status.

Landing Page – This is a page on the school website where students may interact with the page and input their contact information.

Lead – This is a customer contact who has either proactively or re-actively expressed interest in the school through at least one logged point of contact with the school. To be considered a lead, the person during the interaction has input, at the very least, an email address and/or phone number (and optionally, a name) with the school.

New Application Received status – The status of a lead who has completed and submitted an application to the school and paid the application fee. The lead becomes a prospective student.

New Enrolled / Future Start status – The status of a prospective student when they enroll in courses for an upcoming quarter. The prospective student is new to the school and is not reentering a program.

Program – This is the major and degree level of the student. This is synonymous with the data set field progDeg.

Touchpoint – This is a logged point of contact a lead has with the school. All touchpoints by a lead are tracked in an ordered list called a touchpoint trail. Touchpoints are identified by a touchpoint code and the date and time of the point of interaction is recorded. There are many different types of touchpoints ranging from a lead taking a career quiz (touchpoint code CQ) on the school website, to a lead subscribing to a mailing list (touchpoint WCE), hearing a Speaker's Bureau presentation in their high school classroom (SB), or even filling out an online application on the website touchpoint (OA).

CHAPTER 3. METHODOLOGY

At the start of this study the following assumptions were made:

1) Since very little is known about leads other than their touchpoint contacts and their identifying information which is limited to one or more of the following: email addresses, phone numbers and/or first and last names, we assume there are real potential customers behind this information. No internal study has been done in marketing to tease out bogus contacts.
2) The value of a touchpoint in influencing a lead to submit an application and progress to New Applicant Received status remained consistent over the time period (~4.5 years) of data collection. No year-by-year performance breakdown was examined.
3) As in 2), during the modeling stage the likelihoods an applicant will progress to enrolling, and enrollee will progress to active status and an active student will withdraw within the first year are assumed to remain constant over the time period.

The data set of 305,828 observations was derived from joining two data sets in different departments:

4) Marketing department lead tracking database (leads from 06/10/2017 to 09/30/2022)
5) Student advising department student tracking database (prospective students or students from (06/10/2017-12/31/2022)

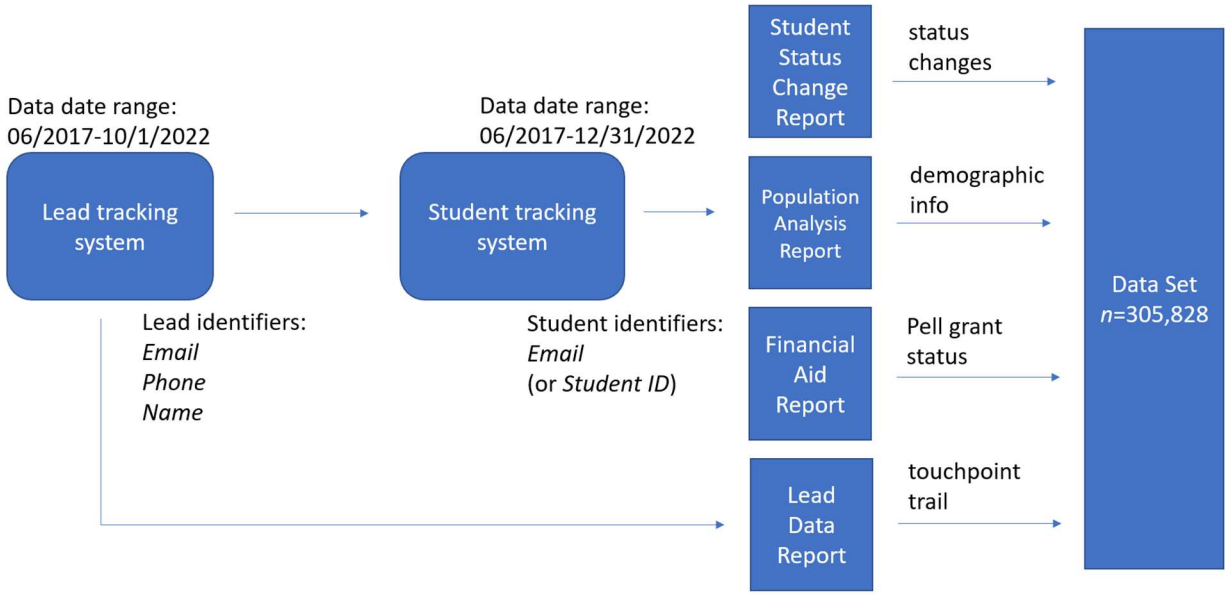

Figure 2. Data Sourcing

Figure 2 describes the sources of data that go into the data set. When a lead interacts with the school via a touchpoint, this observation is logged in the lead tracking database with a time stamp and sequence number. A particular lead may have one to many touchpoints making up a touchpoint trail. There is no upper limit to the number of touchpoints leads may have, but few if any leads in the data had over fifty touchpoints in their trail. If the lead develops into a prospective student by submitting an application, this information is handed off to the student tracking database using email and/or phone number as the primary key. The application requires the prospective student fill out demographic data. Financial data on the prospective student may be included or may be obtained further along in the process. The student tracking system stores all demographic and financial information for the prospective student. The data set for this study used email or phone number (or in rare cases with incomplete information, full name) to *left join* the student tracking data to the lead data, assuming the student tracking data was present.

For most observations in this joined set, the fields for the student tracking database variables will be null because most leads never progress to submitting an application with the school and entering the student tracking database.

The timeline of the data was determined by the date in which the lead tracking system was implemented in 2020 and consists of all leads that had a point of contact with the school from that point forward. The school does not have a data warehouse for immediate access of touchpoint data prior to 2020, so for leads with a preexisting touchpoint trail who had subsequent points of contact post-2020 on, the prior point of contact history is queried via email or phone primary keys and appended to the new touchpoint data.

DATA MANIPULATION/CLEANING - GENERAL

The variables utilized in this study at the broadest level are described in Appendix A.
The following actions on the data set were taken prior to any analysis:

1) All demographic column levels were modified to removes spaces.
2) All date columns were modified to YYYY-MM-DD with hours, minutes, seconds dropped.
3) Touchpoint codes were extracted from the sisCollector column into a new field sisCollectorCode and sisCollector was dropped.
4) Some touchpoints have since been retired and are not relevant to this study. This dropped 5388 observations.
5) There were touchpoints that could be grouped together in a meaningful way. These were grouped under the new umbrella touchpoints:
    - Career Quiz (CQ)
    - Event (E)
    - Fashion Club (FC)
    - Landing Page – Major (LPM)

- Online Application (OA)
- Speaker's Bureau (SB)

6) The major and degree columns do not exist independently, and both need to be known to determine the exact "program" a student is in. The new column prog_deg (called Program in list of variables) was created to concatenate the major and degree column together. Then all programs(Appendix B) were grouped under the following umbrella programs that constitute the three main educational tracks in the school:
   - Apparel (all design programs related to apparel)
   - OtherDesignRelated (all other design programs like Digital Media, Interior Design, etc.
   - Business (all business/merchandising-related programs)

After the above generalized cleaning and manipulation of data, the observations were reduced to N=300,440. Next, the dataset was split into four subsets each addressing the four questions in the broad question as stated in the introduction. All subsets created to answer Questions A), B), C), and D) use this set as the starting point for subsequent sub-setting.

DATA MANIPULATION/CLEANING – QUESTION A)

Question A) What factors are associated with a lead subsequently applying?

The original data set was reduced to only the required columns for analysis of touchpoints and the new column Applied was added.

Variable Name	Type	Description
Email	Nominal	lead
Inq.Create_Date	Date: YYYY-MM-DD	Date lead interacts with touchpoint
Inquiry_Sequence_Number	Ratio (discrete)	The nth inquiry (touchpoint) by a given lead
sisCollectorCode	Nominal	Code representing touchpoint
AppRecDate	Date: YYYY-MM-DD	Date an application was received, and the application fee was paid by a lead. Lead status became New Application Received.

Table 1. Variables for answering Question A)

Outcome Variable Name	Type	Description
Applied	Binary	1 – AppRecDate non-null 0 – otherwise

Table 2. Outcome variable for answering Question A)

As stated before, there was limited information on leads besides the touchpoints they interact with, so the goal of this portion of the study was to tabulate and rank the mean success rates of each touchpoint where a lead applying is considered success. Aside from getting an overview of each touchpoint performance relative to others, marketing was most interested in specific resource intensive touchpoints that will be discussed in the results section.

The following actions on the data set were taken:

1) If we consider the lag time between a lead's first touchpoint (lead first enters the marketing system) and New Application Received status, then it makes sense to subset the data according to a cutoff date for allowing new leads with first-touchpoints into the subset. This will give the most recently recorded leads an average amount of time to progress to the next status, if at all before 12/31/2022, when the data set ends. The mean lag time for leads from first touchpoint to New Application Received status was 171 days, while the median was 61 days with a roughly 120-day in-between point which will be used here as the cutoff. Figure 3 illustrates the distribution of lag times in days from first touchpoint (Inq.CreateDate where Inquiry_sequence_number == 1) to submitting an application (AppRecDate). Using 4 months as a guide, all leads with first touchpoints after 08/30/2022 were dropped.

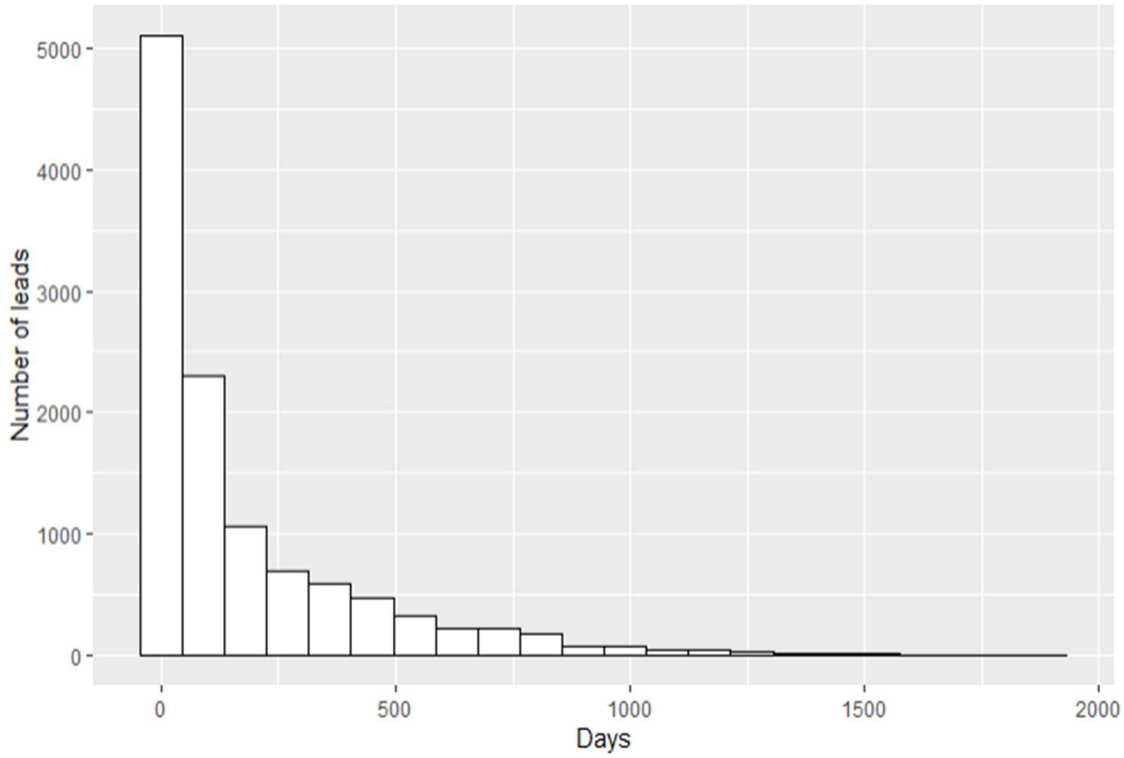

Figure 3. Days from first touchpoint to New Application Received status

2) It is not uncommon for a lead to have multiple instances of the same touchpoint in their touchpoint trail. For instance, the lead may take the Career Quiz (CQ) multiple times just out of curiosity. This study will eliminate all but the most current duplicate touchpoint in each lead's touchpoint trail, treating the most current as the "freshest".
3) All touchpoints a lead had with Inq.CreateDate after the date of the status change to New Application Received (AppRecDate) were dropped.

The resulting data set for answering Question A) had N_{lead}=240,512 observations.

DATA MANIPULATION/CLEANING – QUESTION B)

Question B) What factors are associated with an applicant progressing to enrollee?

The following set of predictor variables were used:

Predictor Variable Name	Type	Description
Program	Nominal	Program-type student is pursuing. - Apparel - Business - Other Design Related
AgeBracket	Ordinal	Age range of student - Under 18 - 18-19 - 20-21 - 22-24 - 25-29
Gender	Nominal	Gender of student - Female - Male
Ethnicity	Nominal	Ethnicity of student / or International if foreign student - American Indian or Alaska Native - Asian - Black or African American - Decline to Respond - Hispanic (Hispanic / Latino) - International - Native Hawaiian or Other Pacific Islander - Two or More Races - White
Pell	Binary	Did prospective student receive a Pell grant? - Yes - No
hasCH	Binary	Was CH in touchpoint trail? - Yes - No
hasFC	Binary	Was FC in touchpoint trail? - Yes - No
hasSB	Binary	Was SB in touchpoint trail? - Yes - No

Table 3. Variables for answering Questions B), C), D)

The outcome variable was created referencing both the EnrollDate field and AppRecDate field per Table A1 in Appendix A.

Outcome Variable Name	Type	Description
enrolled	Binary	Applicant enrolled Yes – EnrollDate is non-null and not less than AppRecDate No – otherwise

Table 4. Outcome variable for answering Question B)

Additional steps taken for answering Question B):

1) All leads that did not apply (AppRecDate was N/A) were dropped from the set.
2) All observations of a touchpoint that occurred after a lead reached New Enrolled / Future Start status were dropped.
3) Since the object of interest in this section is the prospective student, and the original data, if we recall from Section 1), could have multiple observations per lead (now a prospective student), each corresponding to touchpoint event in their touchpoint trail, the observations needed to be grouped by prospective student and collapsed to a single observation. Each touchpoint trail was encoded and appended to the prospective student observation. The result was a data set with only unique prospective student observations.
4) Some high resource touchpoints of particular interest were CH, FC, and SB and in this section, we will check their impact in relation to the demographic and financial factors in our expanded data set. Three new variables hasCH, hasFC, and hasSB were created to check if any of these touchpoints were in the prospective student's touchpoint trail.
5) Before modeling, we looked at the rates of (prospective) students graduating from one status to the next. Considering the data set ends at 12/31/2022, a cutoff date for new data was established with respect to the average lag time between status changes from New Application Received to New Enrolled / Future Start. All applicants with application dates after the cutoff were dropped. The question arose as to whether there would be different lag times between the group of prospective students in the Under18 group and

everyone else. The Under18 group is still in high school when applying so it is not unreasonable to assume lag time from New Application Received status to New Enrolled / Future Start status could be different than for everyone else. Comparing distributions of lag times between the two groups does not show much difference:

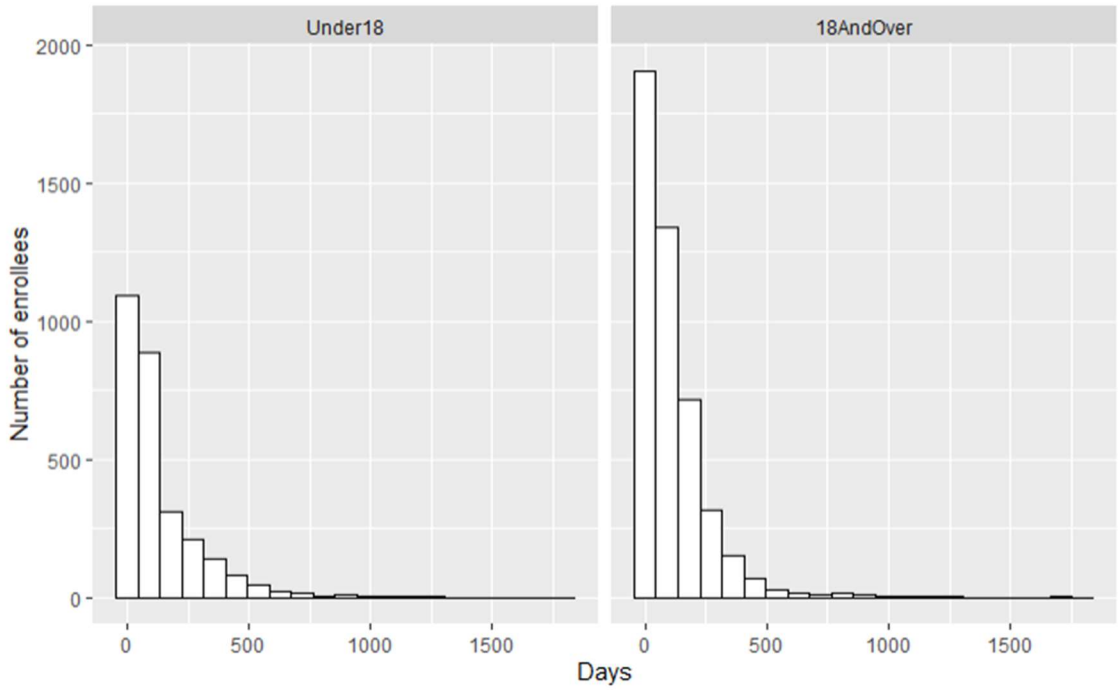

Figure 4. Lag time in days from application to enrollee status by age bracket

Therefore, we used the same cutoff between the groups. The overall mean lag time was 122 days with a median of 63 days. A 90-day lag time was used and all applicants after 9/30/2022 were dropped.

6) Finally, one observation of “Not Specified” for Gender was dropped.

The resultant data set after processing was a pool of applicants with $N_{applicant}$=15,443. Of these, 6,356 enrolled (progressed to New Enrolled / Future Start status). This is a 59% drop in enrollments from applicants.

EXPLORATORY DATA ANALYSIS – QUESTION B)

Prior to analysis of the variables, the data was split 80%/20% into training and testing sets respectively. All subsequent analysis was on the training set.

An examination of the variables indicate that each variable level has sufficient data for analysis, but in some cases, there were too many levels with low frequencies that proved insignificant during preliminary modeling. Under Ethnicity, the American Indian / Alaskan Native and Native Hawaiian / Pacific Islander levels were merged under IndigAmerican, short for Indigenous American. Also, an examination of incoming student ethnicities (self-reported by the prospective student) and their ethnicities as established students after follow-up by Student Advising staff show that the vast majority of those who declared ethnicity of Two or more races were later clarified as Hispanic. Therefore, these two groups were merged into Hispanic.

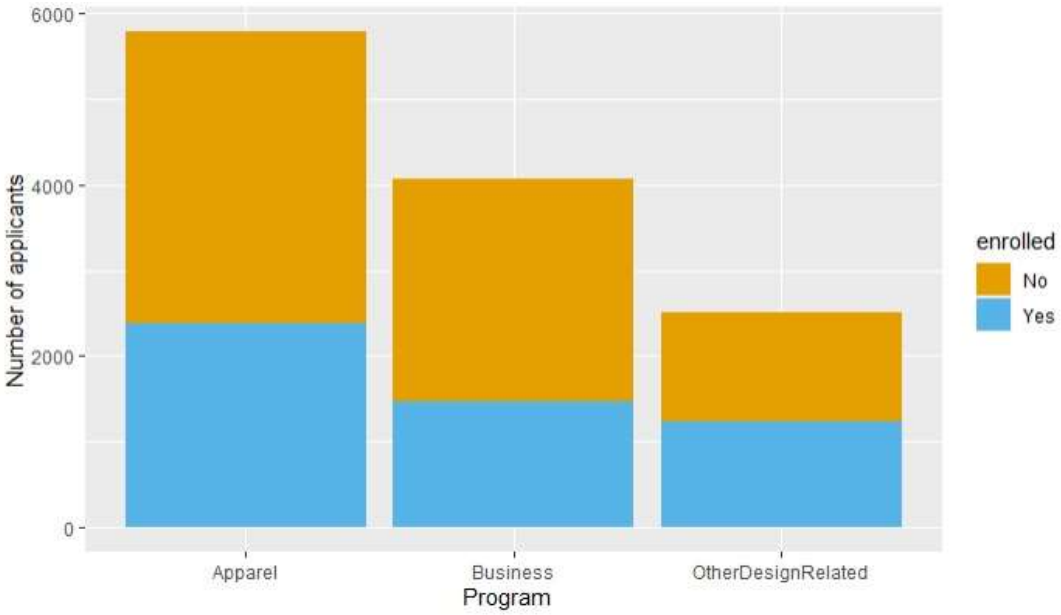

Figure 5. Enrollment by Program

Enrolled	Apparel	Business	OtherDesignRelated
No	3411	2593	1266
Yes	2374	1474	1237

Table 5. Enrollment by Program

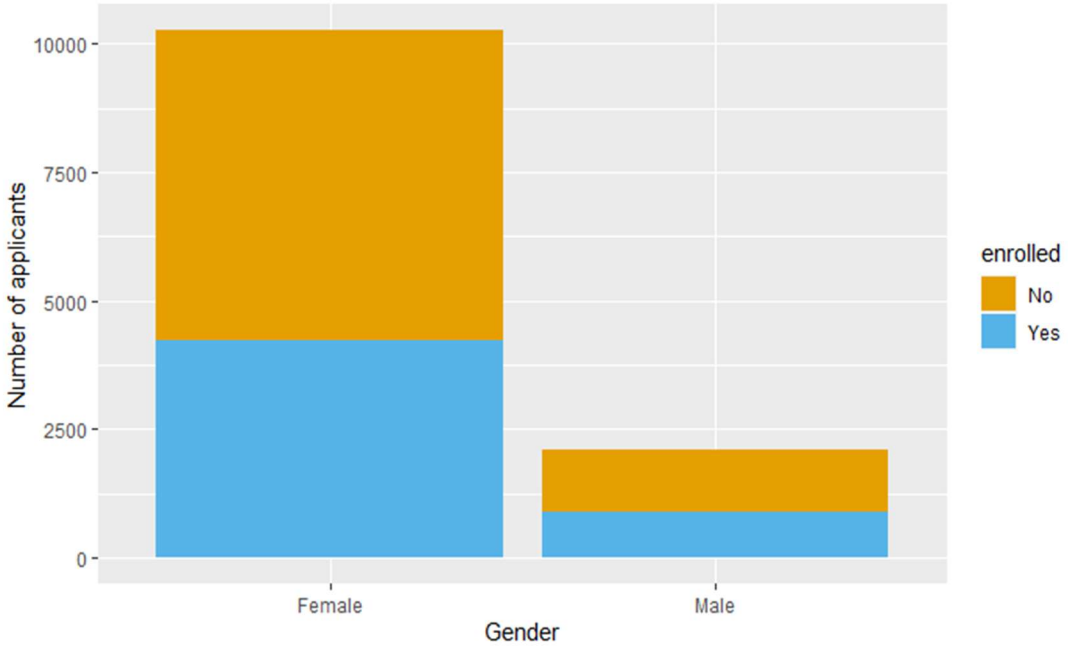

Figure 6. Enrollment by Gender

Enrolled	Female	Male
No	6048	1222
Yes	4207	878

Table 6. Enrollment by Gender

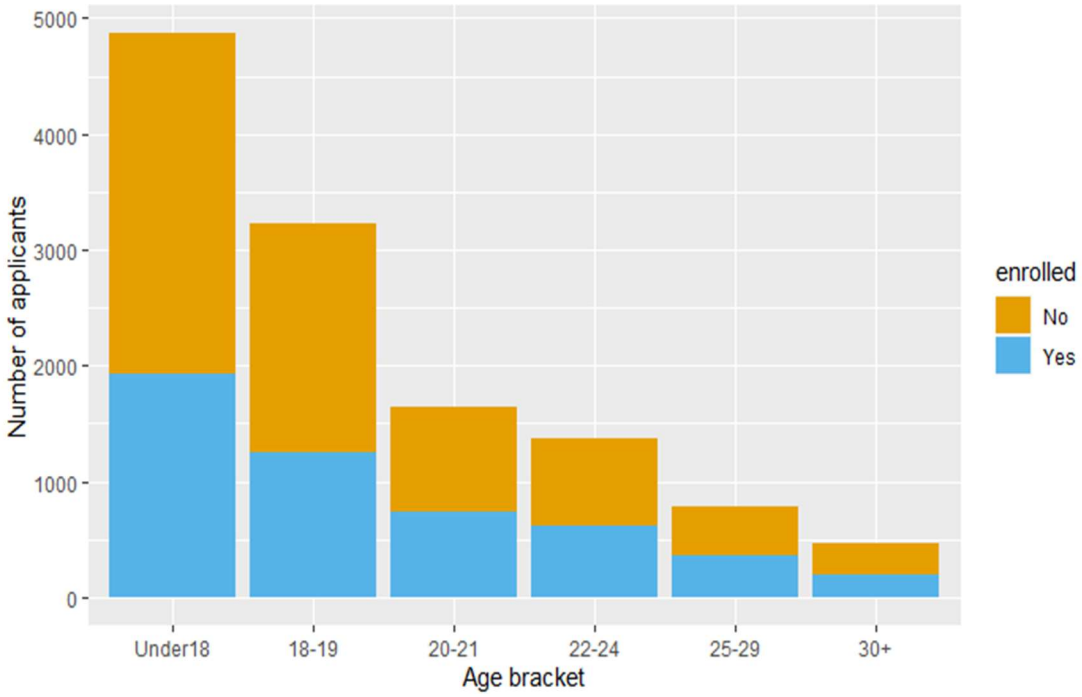

Figure 7. Enrollment by Age bracket

Enrolled	Under18	18-19	20-21	22-24	25-29	30+
No	2933	1979	897	753	432	276
Yes	1934	1254	741	613	353	190

Table 7. Enrollment by Age bracket

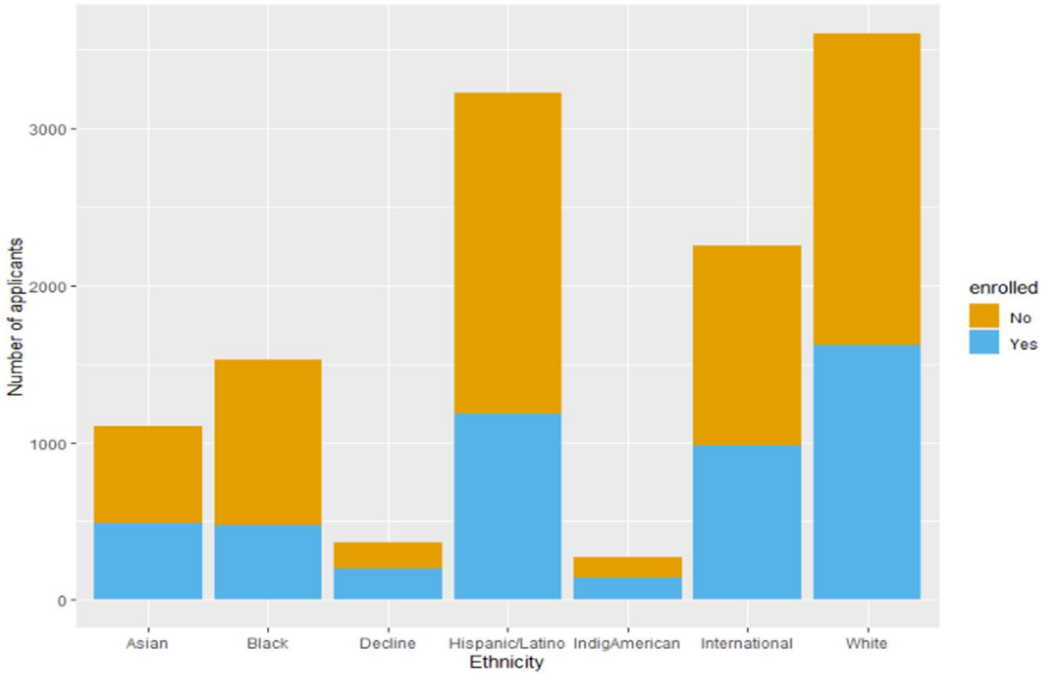

Figure 8. Enrollment by Ethnicity

Enrolled	Asian	Black	Decline	Hispanic	IndigAmerican	International	White
No	619	1053	168	2042	133	1272	1983
Yes	489	471	197	1186	139	981	1622

Table 8. Enrollment by Ethnicity

IndigAmerican is short of Indigenous American and resulted from merging the categories American Indian or Alaska Native and Native Hawaiian or Pacific Islander, both of which had very low frequencies.

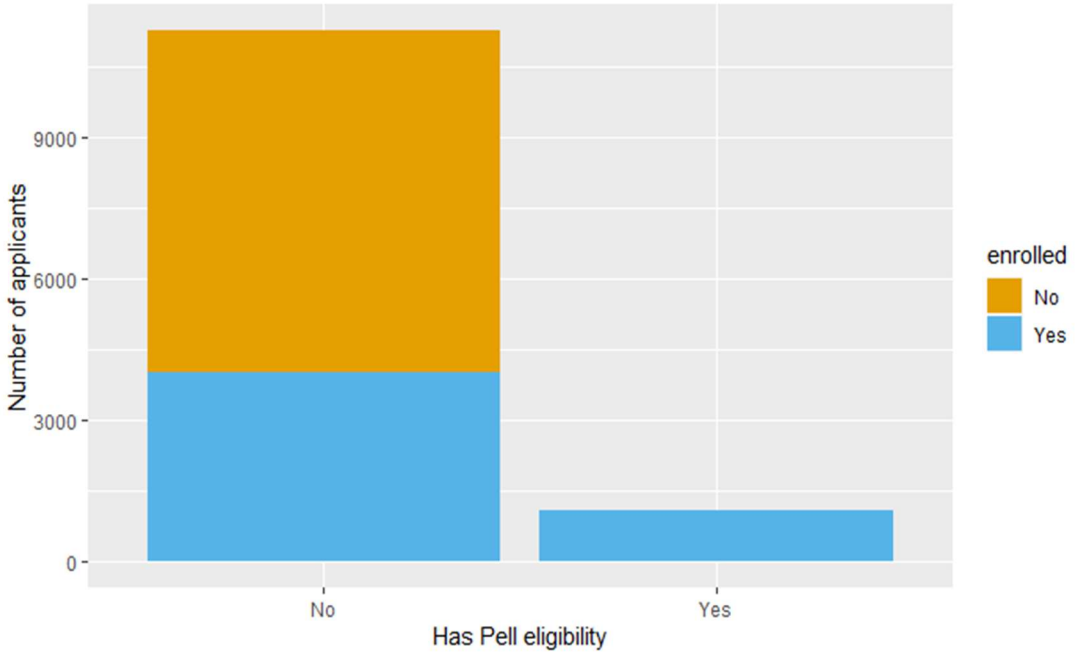

Figure 9. Enrollment by Pell eligibility

Enrolled	No	Yes
No	7250	20
Yes	4022	1063

Table 9. Enrollment by Pell eligibility

This study was unable to identify the date that Pell eligibility was input into the student tracking database. The distribution in Figure 9 suggests that it was almost always after the student had at the very least progressed to enrollee status If an applicant cancelled their application and did not progress further, there would be no record of their Pell eligability status being updated in the affirmative. Therefore, it is likely Pell status is conditional on the applicant progressing to enrollee status. Due to this issue, Pell is not a reliable predictor for use in modeling the likelihood of applicants progressing to enrollee status.

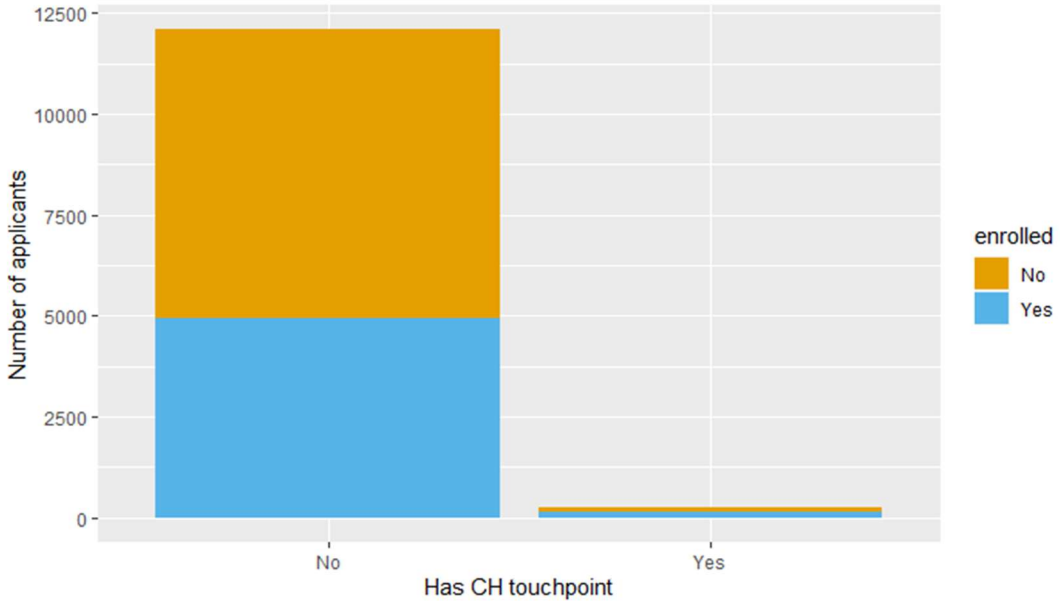

Figure 10. Enrollment when CH touchpoint in touchpoint trail

Enrolled	No	Yes
No	7140	130
Yes	4945	140

Table 10. Enrollments when CH touchpoint in touchpoint trail

The CH touchpoint represents use of the chat functionality on the school website. The inclusion of this variable (whether the applicant had at some time used the chat functionality on the school website) as a modeling variable was part of an attempt to check the its power in driving enrollment. 2.7% of those applicants with the chat touchpoint enrolled verses 1.8% who did not have the touchpoint. Of course other factors could have been involved in this percent difference such as greater interest / engagement to begin with in those that utilized the chat functionality.

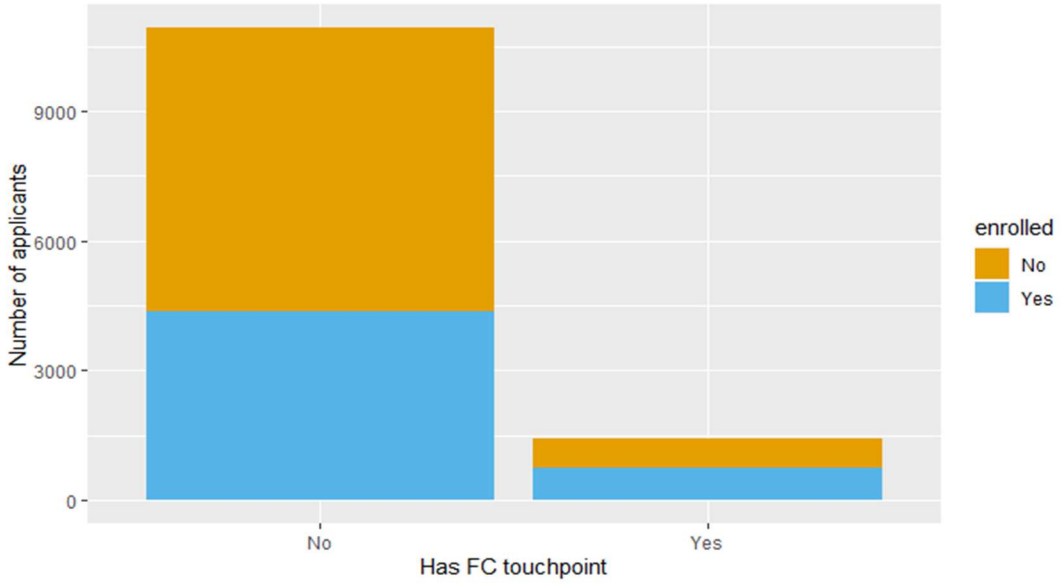

Figure 11. Enrollment when FC touchpoint in touchpoint trail

Enrolled	No	Yes
No	6572	698
Yes	4350	735

Table 11. Enrollment when FC touchpoint in touchpoint trail

The FC touchpoint represents participation in the school sponsored fashion club. The inclusion of this variable as a modeling variable was part of an attempt to check the its power in driving enrollment. 14.5% of those with the fashion club touchpoint enrolled verses 9.6% of those without that touchpoint. Of course other factors could have been involved in this percent difference such as greater interest / engagement to begin with in those that utilized the chat functionality.

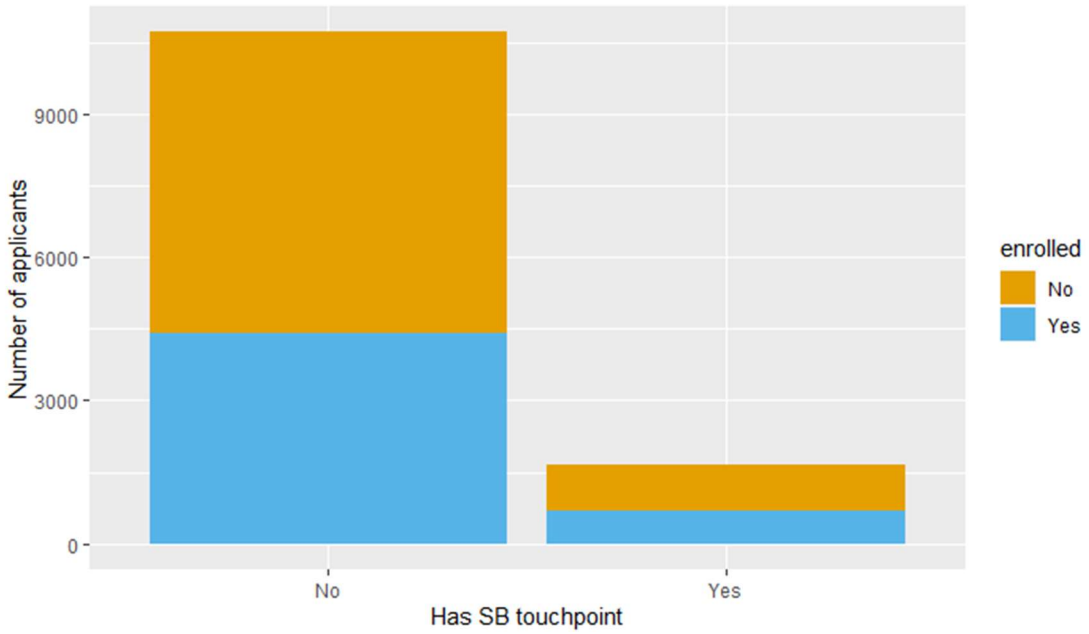

Figure 12. Enrollment when SB touchpoint in touchpoint trail

Enrolled	No	Yes
No	6314	956
Yes	4399	686

Table 12. Enrollments when SB touchpoint in touchpoint trail

The SB touchpoint represents contact with the speaker's bureau. The inclusion of this variable as a modeling variable was part of an attempt to check the its power in driving enrollment. 13.5% of those with the speaker's bureau touchpoint enrolled verses 13.1% of those without that touchpoint. This suggests the touchpoint SB is not a significant driver of enrollment.

DATA MANIPULATION/CLEANING – QUESTION C)

Question C) What factors are associated with an enrollee progressing to active student status?

The predictor variables are identical to those in Table 4, now in the service of answering Question C).

The outcome variable was created referencing both the ActiveDate field and EnrollDate field per Table A1 in Appendix A.

Outcome Variable Name	Type	Description
active	Binary	Enrollee progressed to Active status Yes – ActiveDate is non-null and not less than EnrollDate No – otherwise

Table 13. Outcome variable for answering Question C)

Additional steps taken for answering Question C):

1) All applicants that did not enroll (EnrollDate was N/A) were dropped from the set.
2) All observations of a touchpoint that occurred after an enrollee reached Active status were dropped.
3) See 3) under Additional steps taken for answering Question B).
4) See 4) under Additional steps taken for answering Question B).
5) The logic in this step progressed in the same manner as in 5) under Additional steps taken for answering Question B). When taking into account the lag time from New Enrolled / Future Start status to Active status and determining the optimal cutoff date an examination of the distribution of prospective students still in high school and all other prospective students was done.

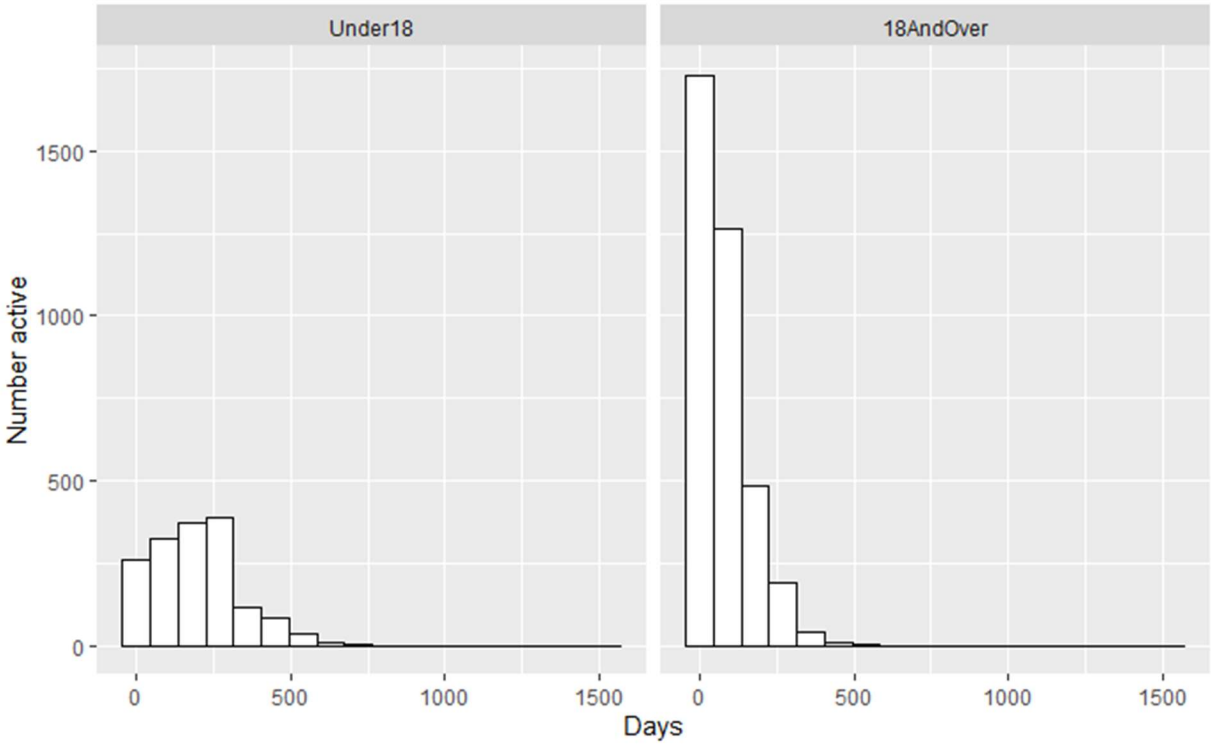

Figure 13. Lag time in days from enrollee to active student status by Age bracket

Because of the different distributions, this study used different cutoffs for each age group. The mean lag time for Under18 was 197 days with a median of 194 and the mean lag time for the 18AndOver was 78 with a mean of 51. For the Under18 group we used 210 days and for the 18AndOver group we used 90 days.

The resultant data set after processing was a pool of enrollees with $N_{enrollee}$=6,955. Of these, 4,648 progressed to Active status. This is a 33% drop in students progressing to Active status from New Enrolled / Future Start status.

EXPLORATORY DATA ANALYSIS – QUESTION C)

Prior to analysis of the variables, the data was split 80%/20% into training and testing sets respectively. All subsequent analysis was on the training set.

As in the analysis for Question B), the ethnic variable levels of American Indian / Alaskan Native and Native Hawaiian / Pacific Islander were merged into IndigAmerican. because of the relatively low frequencies. Additionally, the level Two or more races was merged with Hispanic.

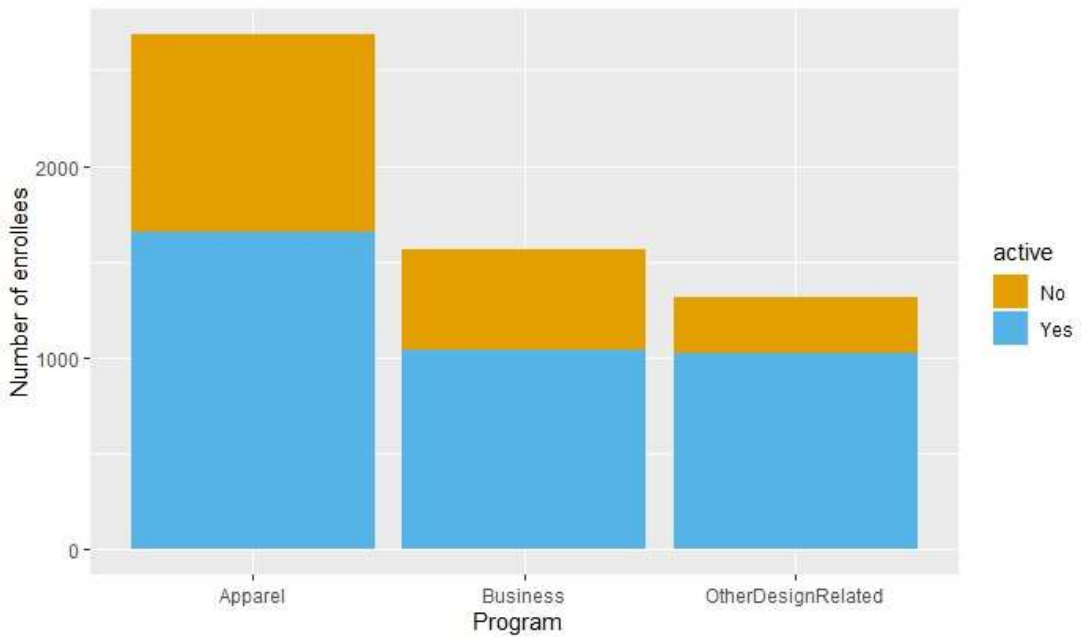

Figure 14. Progression to Active status by Program

Active	Apparel	Business	OtherDesignRelated
No	1028	521	297
Yes	1657	1043	1019

Table 14. Progression to Active status by Program

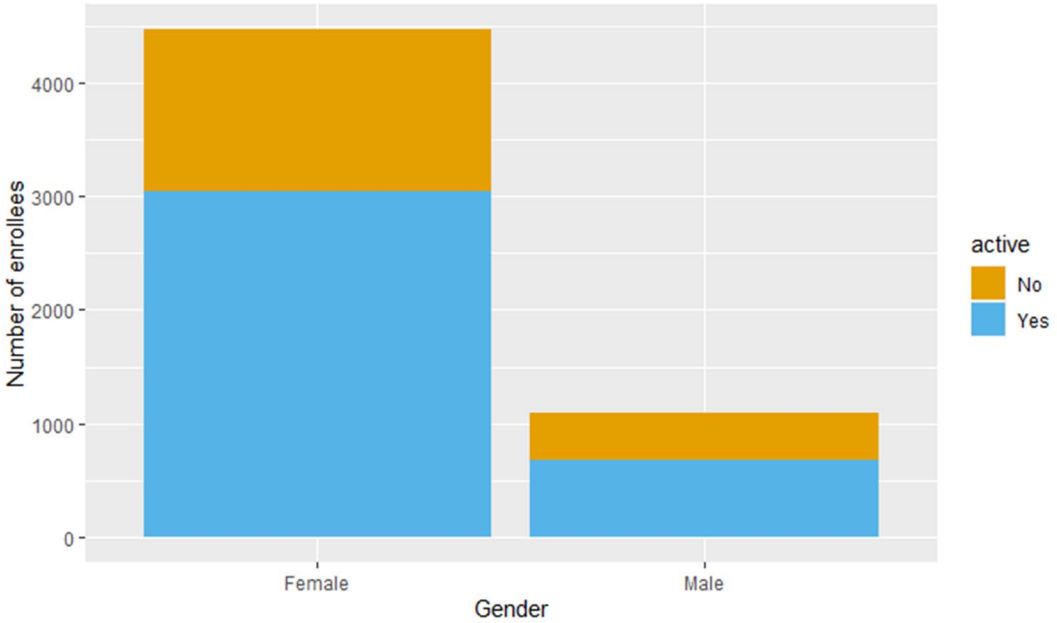

Figure 15. Progression to Active status by Gender

Active	Female	Male
No	1422	424
Yes	3048	671

Table 15. Progression to Active status by Gender

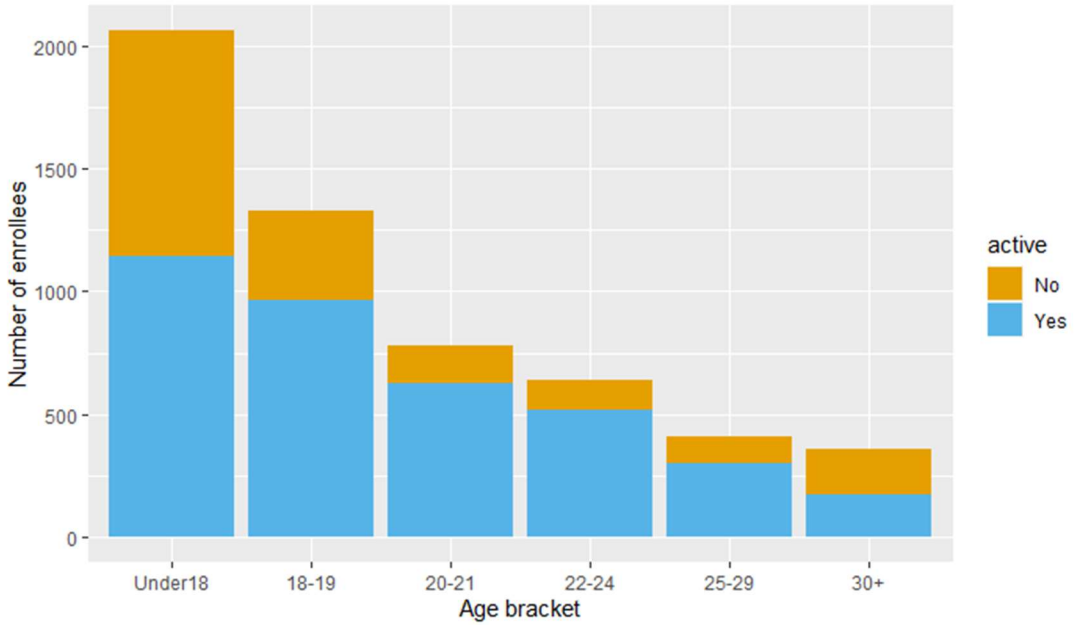

Figure 16. Progression to Active status by Age bracket

Active	Under18	18-19	20-21	22-24	25-29	30+
No	920	363	153	121	106	183
Yes	1143	964	623	517	300	172

Table 16. Progression to Active status by Age bracket

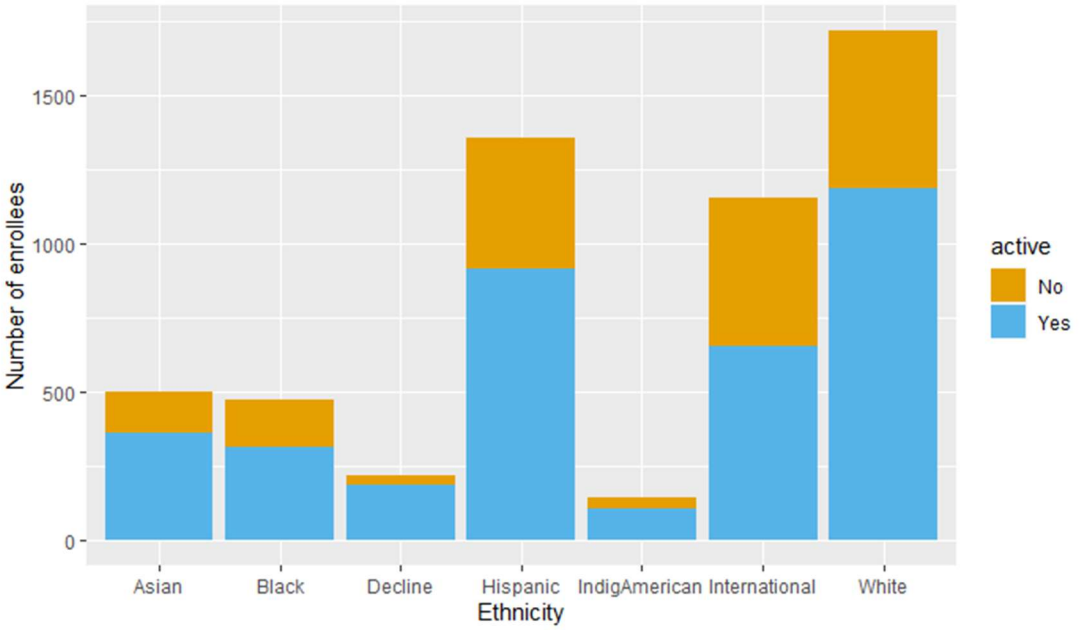

Figure 17. Progression to Active status by Ethnicity

Active	Asian	Black	Decline	Hispanic	IndigAmerican	International	White
No	138	158	36	444	37	501	532
Yes	363	314	183	912	107	655	1185

Table 17. Progression to Active status by Ethnicity

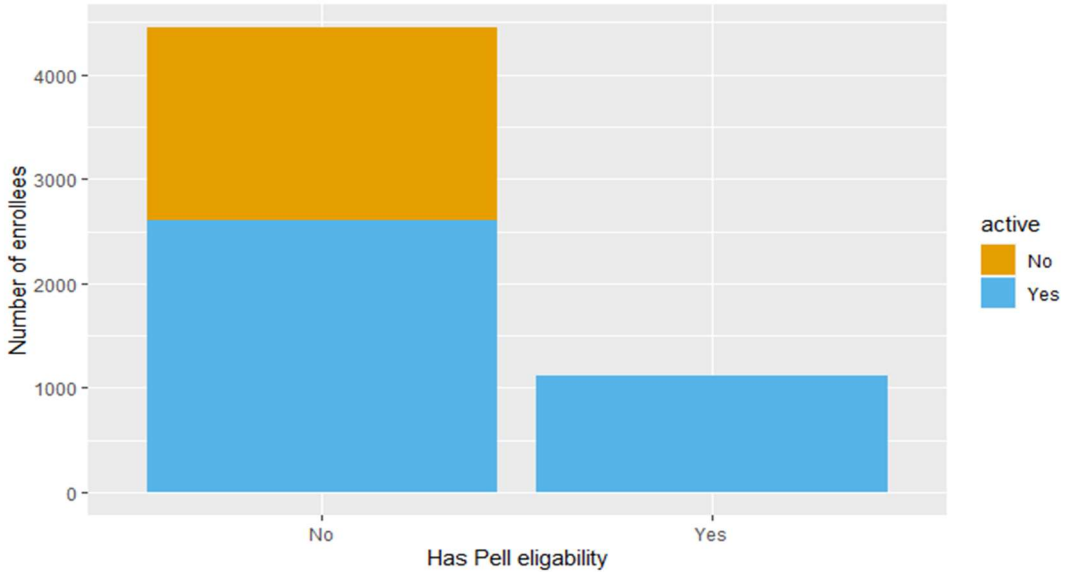

Figure 18. Progression to Active status by Pell eligibility

Active	No	Yes
No	1844	2
Yes	2604	1115

Table 18. Progression to Active status by Pell eligibility

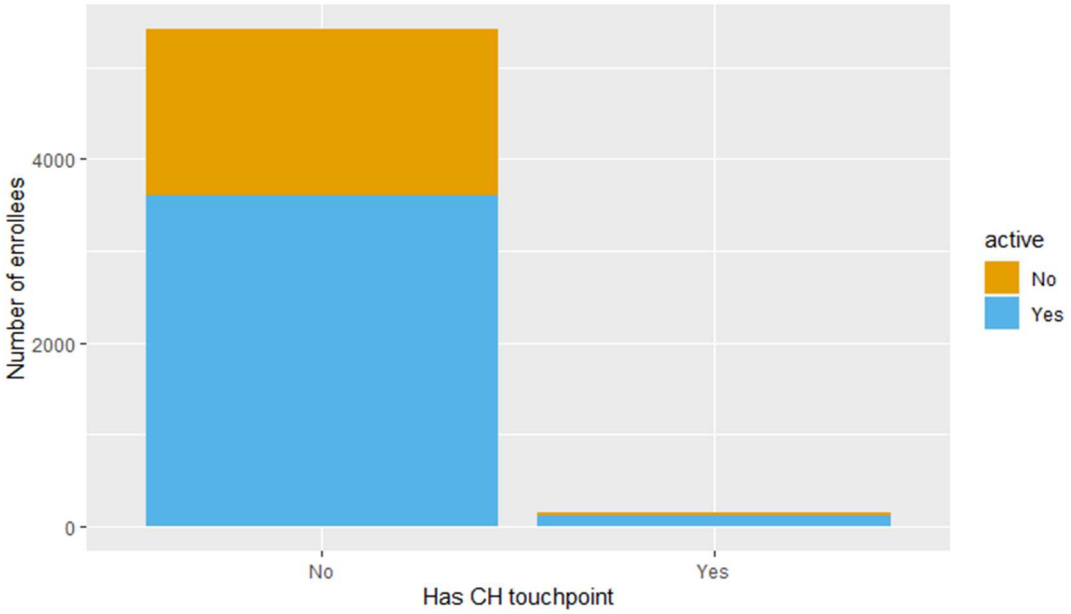

Figure 19. Progression to Active status when CH touchpoint in touchpoint trail

Active	No	Yes
No	1809	37
Yes	3604	115

Table 19. Progression to Active status when CH touchpoint in touchpoint trail

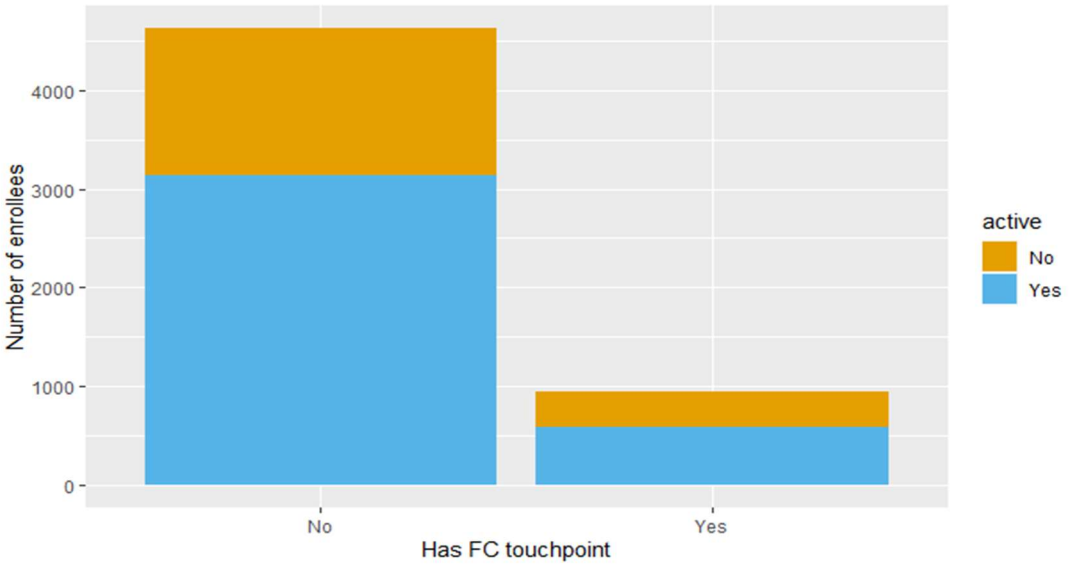

Figure 20. Progression to Active status when FC touchpoint in touchpoint trail

Active	No	Yes
No	1489	357
Yes	3137	582

Table 20. Progression to Active status when FC touchpoint in touchpoint trail

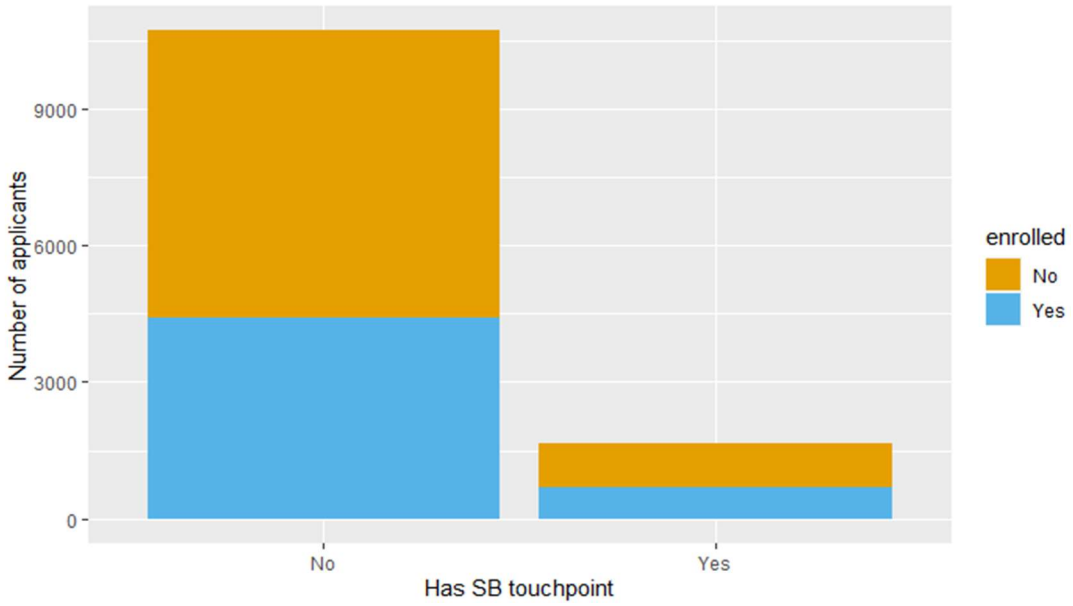

Figure 21. Progression to Active status when SB touchpoint in touchpoint trail

Active	No	Yes
No	1509	337
Yes	3292	427

Table 21. Progression to Active status when SB touchpoint in touchpoint trail

DATA MANIPULATION/CLEANING – QUESTION D)

Question D) What factors are associated with an active student withdrawing within the first year?

The following set of predictor variables were used:

Predictor Variable Name	Type	Description
Program	Nominal	Program-type student is pursuing. - Apparel - Business - Other Design Related
AgeBracket	Ordinal	Age range of student - Under 18 - 18-19 - 20-21 - 22-24 - 25-29

		- 30+
Gender	Nominal	Gender of student - Female - Male
Ethnicity	Nominal	Ethnicity of student / or International if foreign student - American Indian or Alaska Native - Asian - Black or African American - Decline to Respond - Hispanic (Hispanic / Latino) - International - Native Hawaiian or Other Pacific Islander - Two or More Races - White
Pell	Binary	Did prospective student receive a Pell grant? - Yes - No

Table 22. Outcome variables for answering Question D)

The outcome variable was created referencing both the WithdrawalDate field and ActiveDate field per Table A1 in Appendix A.

The outcome variable:

Outcome Variable Name	Type	Description
withdrawal	Binary	Active student withdrew within 1 year Yes – WithdrawalDate is non-null, not less than ActiveDate, and not greater than 365 days from ActiveDate No – otherwise

Table 23. Outcome variables for answering Question D)

Additional steps taken for answering Question D):

1) All enrollees who did not become active students (ActiveDate was N/A) were dropped from the set.
2) One observation of “Not Specified” for Gender was dropped.

Of note here, the variables tied to touchpoints: hasCH, hasFC and hasSB are irrelevant at this stage of a student’s progress considering they are active students.

The resultant data set after processing was a pool of active students with N_{active}=4554. Of these, 1130 withdrew within the first year. This is a 25% withdraw rate within the first year of attendance.

EXPLORATORY DATA ANALYSIS – QUESTION D)

Prior to analysis of the variables, the data was split 80%/20% into training and testing sets respectively. All subsequent analysis was on the training set.

As in the analysis for Questions B) and C), the ethnic variable levels of American Indian / Alaskan Native and Native Hawaiian / Pacific Islander were merged into IndigAmerican. because of the relatively low frequencies. Additionally, the level Two or more races was merged with the level Hispanic.

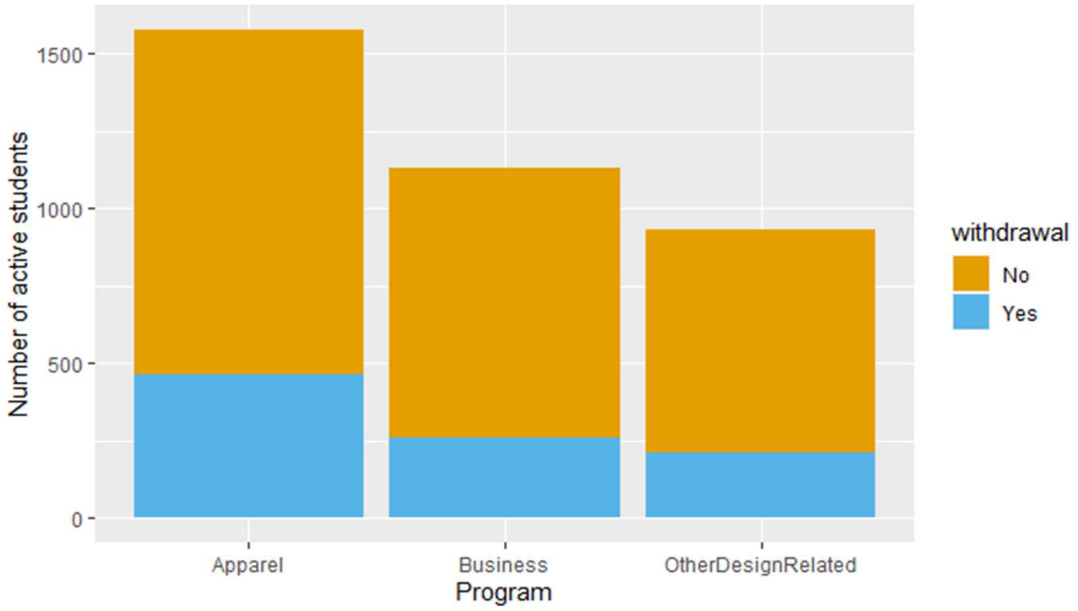

Figure 22. Withdrawal within one year by Program

Withdrawal within one year	Apparel	Business	OtherDesignRelated
No	1129	882	729
Yes	439	266	199

Table 24. Withdrawal within one year by Program

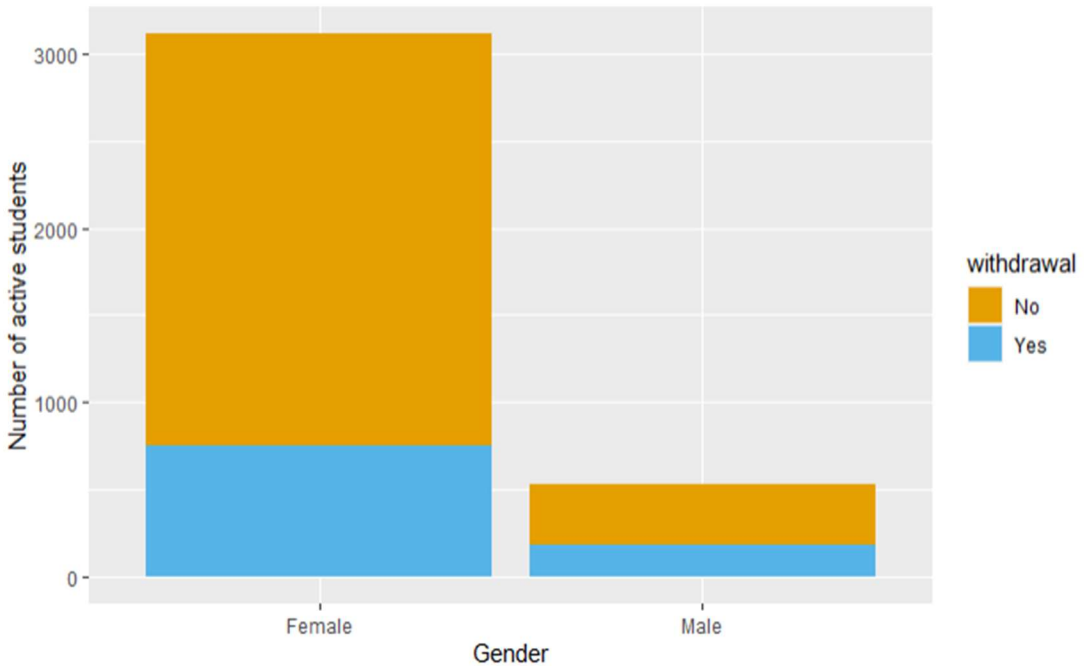

Figure 23. Withdrawal within one year by Gender

Withdrawal within one year	Female	Male
No	2394	346
Yes	723	181

Table 25. Withdrawal within one year by Gender

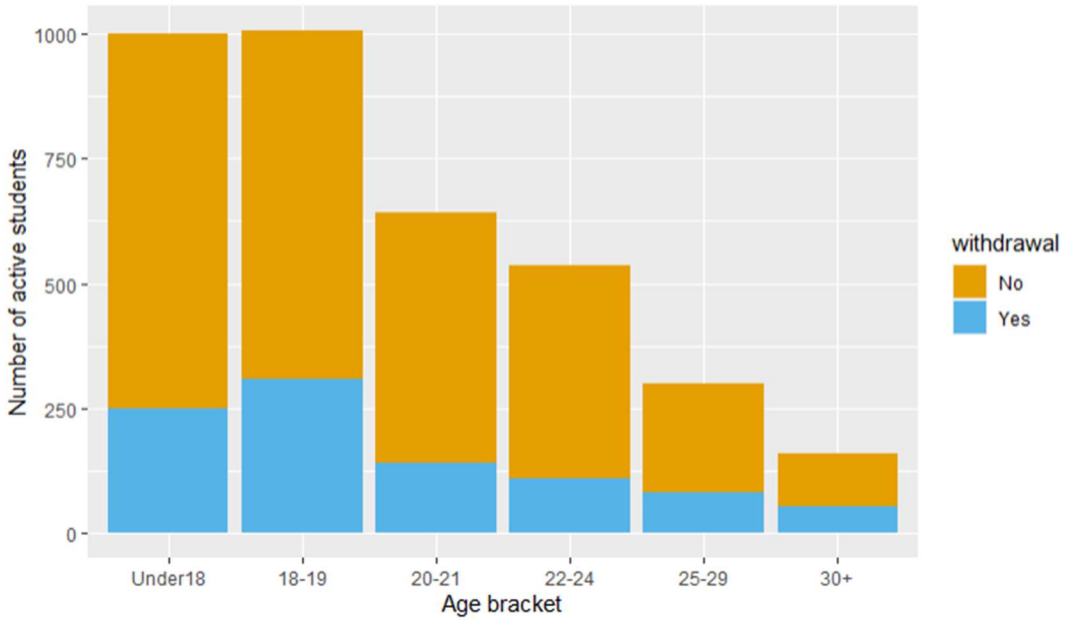

Figure 24. Withdrawal within one year by Age bracket

Withdrawal within one year	Under18	18-19	20-21	22-24	25-29	30+
No	787	688	506	433	227	99
Yes	240	315	135	91	75	48

Table 26. Withdrawal within one year by Age bracket

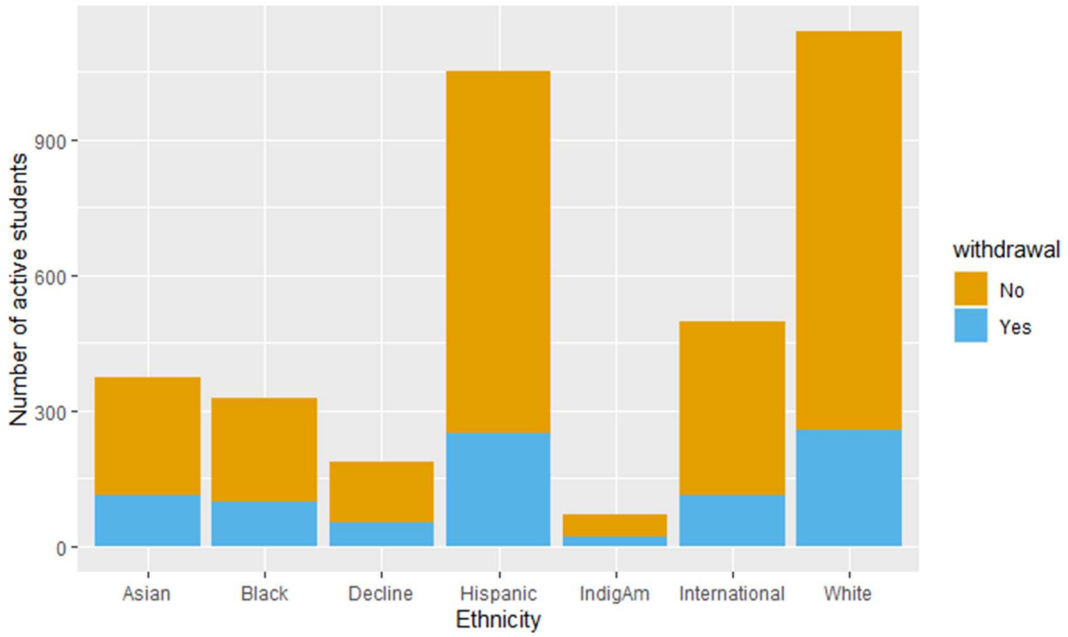

Figure 25. Withdrawal within one year by Ethnicity

Withdrawal within one year	Asian	Black	Decline	Hispanic	IndigAmerican	International	White
No	261	229	135	800	49	384	882
Yes	111	100	51	251	22	112	257

Table 27. Withdrawal within one year by Ethnicity

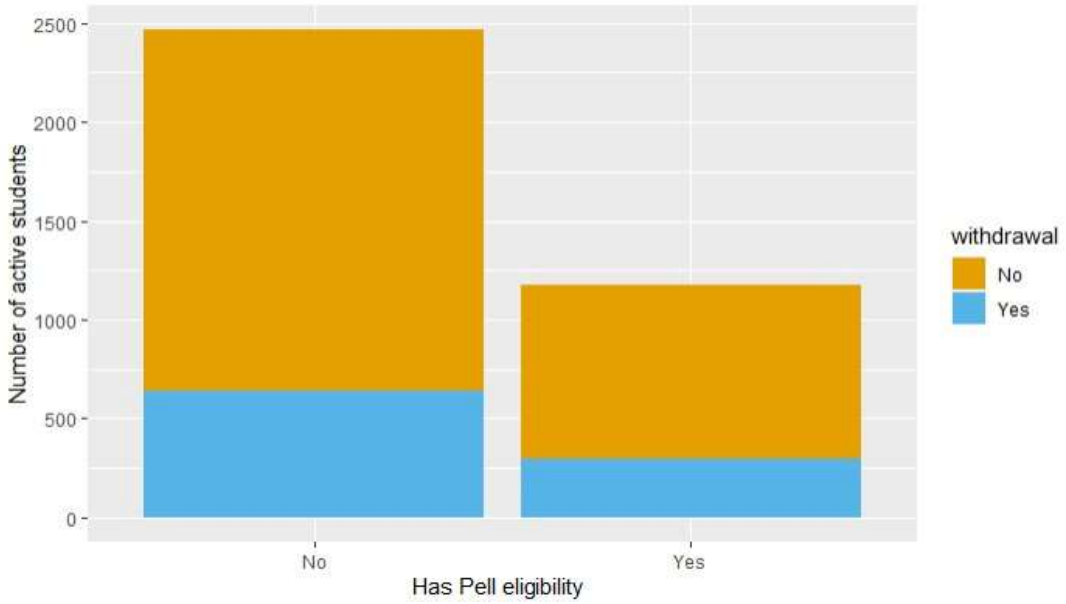

Figure 26. Withdrawal within one year by Pell eligibility status

Withdrawal within one year	No	Yes
No	1855	885
Yes	616	288

Table 28. Withdrawal within one year by Pell eligibility status

CHAPTER 4. RESULTS

All estimates used a 0.05 level of significance.

Results when answering Question A) were estimations of the likelihood a touchpoint would be associated with a lead that subsequently applied. These estimations were done independently for each touchpoint that existed in the data and were ranked in descending order of likelihood. Along the way, five touchpoints were removed:

- LTR was never associated with a lead that subsequently applied.
- R had too few observations to use a normal approximation to the binomial distribution when using criteria np, n(1-p) ≥ 5.
- FDB, TIR, and WSL had lower margins of error that crossed 0.

See Table B1 for more information on the above touchpoints.

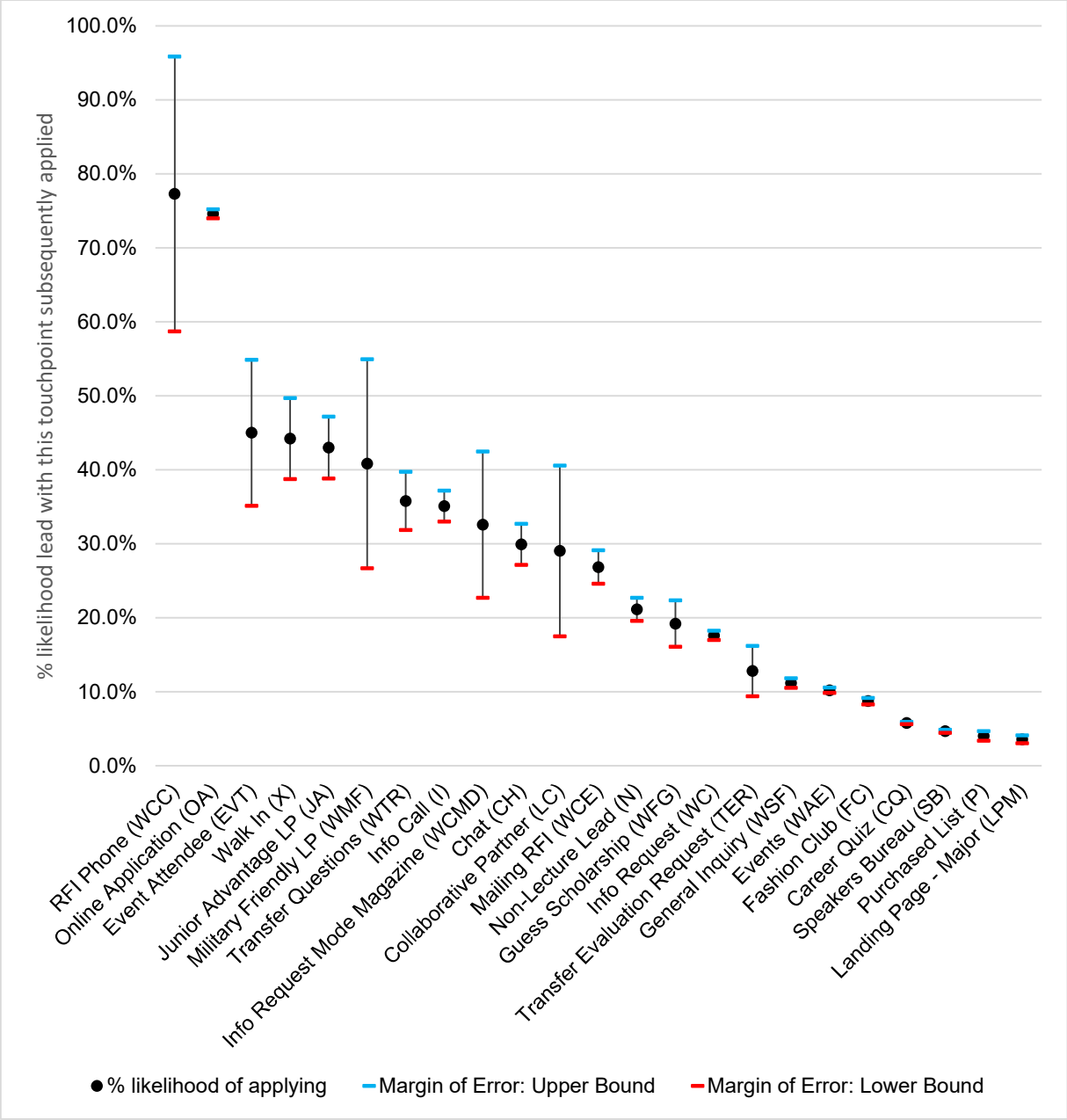

Figure 27. Percent likelihood touchpoint is associated with a lead that subsequently applies

Touchpoint	Did not apply	Applied	Total touchpoint occurrences	Probability of applying	Margin of Error: Lower Bound	Margin of Error: Upper Bound
RFI Phone (WCC)	5	17	22	77.3%	58.7%	95.9%
Online Application (OA)	4821	14155	18976	74.6%	74.0%	75.2%
Event Attendee (EVT)	55	45	100	45.0%	35.1%	54.9%
Walk In (X)	178	141	319	44.2%	38.7%	49.7%
Junior Advantage LP (JA)	309	233	542	43.0%	38.8%	47.2%
Military Friendly LP (WMF)	29	20	49	40.8%	26.7%	54.9%
Transfer Questions (WTR)	368	205	573	35.8%	31.8%	39.7%
Info Call (I)	1286	695	1981	35.1%	33.0%	37.2%
Info Request Mode Magazine (WCMD)	60	29	89	32.6%	22.7%	42.5%
Chat (CH)	736	314	1050	29.9%	27.1%	32.7%
Collaborative Partner (LC)	44	18	62	29.0%	17.5%	40.6%
Mailing RFI (WCE)	1076	395	1471	26.9%	24.6%	29.1%
Non-Lecture Lead (N)	2096	562	2658	21.1%	19.6%	22.7%
Guess Scholarship (WFG)	492	117	609	19.2%	16.1%	22.3%
Info Request (WC)	11213	2398	13611	17.6%	17.0%	18.3%
Transfer Evaluation Request (TER)	327	48	375	12.8%	9.4%	16.2%
General Inquiry (WSF)	8053	1012	9065	11.2%	10.5%	11.8%
Events (WAE)	24788	2817	27605	10.2%	9.8%	10.6%
Fashion Club (FC)	14339	1371	15710	8.7%	8.3%	9.2%
Career Quiz (CQ)	93265	5724	98989	5.8%	5.6%	5.9%
Speakers Bureau (SB)	36661	1796	38457	4.7%	4.5%	4.9%
Purchased List (P)	3361	141	3502	4.0%	3.4%	4.7%
Landing Page - Major (LPM)	4529	168	4697	3.6%	3.0%	4.1%
Total	208091	32421	240512	13.5%		

Table 29. Percent likelihood touchpoint is associated with a lead that subsequently applies

Results when answering Question B) utilized the expanded set of variables that were available. Recall that at this stage in the Status Pathway for Students (Figure 1), a student has applied to the school and their application form will contain demographic information, along with, potentially, some limited financial data. We decided to use logistic regression due to its ease of interpretation, and our large sample size would allow for it ($N_{applicant}$=15,443, with 6,356 going on to enroll).

The modeling strategy was backwards elimination because, going into this study, there were no preconceptions beyond anecdotes as to how these variables would relate with an applicant moving to New Enrolled / Future Start status. The original plan was to incorporate the variable Pell into the modeling because it was the one variable that acted as a proxy for financial status. However, during the data collection process, no time stamps for when Pell eligibility status was determined were available. Because of this instability, we can image situations where an applicant does not progress to New Enrolled / Future Start status and drops off before Pell eligibility was determined. In this case, the default answer for eligibility would be “no”, because no record exists, even if the applicant was in fact eligible. For this reason, Pell status will not be included in modeling.

Model B1: The first model was the full model and most AgeBracket levels were not significant. Student Advising suggested we can make some reasonable assumptions about the Under18 group to separate them from the other age brackets. All prospective students in the Under18 group are unified in they are still in high school and would most likely be entering as first-time, full-time students pursuing an AA. We can make far fewer assumptions about the 18 and over group. Accordingly, the age brackets levels 18-19, 20-21, 22-24, 25-29, 30+ were merged into 18AndOver. In ethnicity IndigAmerican was not significant. We decided to group this level with Hispanic under HispIndig with the advice of the student advising department. While this linking together is not ideal, the IndigAmerican group sample size was too small to stand on its own.

The level White was also not significant. The baseline variable level is the earliest in the alphabet. In this case, under the Ethnicity variable, it was Asian, so we grouped the levels Asian and White under AsianWhite. International remained insignificant so this was also grouped with AsianWhite as AsianIntWhite. At this point, only gender and hasSB were insignificant. Since, hasSB was ranked worse, it was dropped for Model B2.

Model B2: All variables were significant but Gender. It was removed for Model B3.

Model B3: All variables were significant. In all, the AIC went from 16396 in the full model to 16393 in the final model, a very small change, and an ANOVA chi-square test between the three models showed no significant differences.

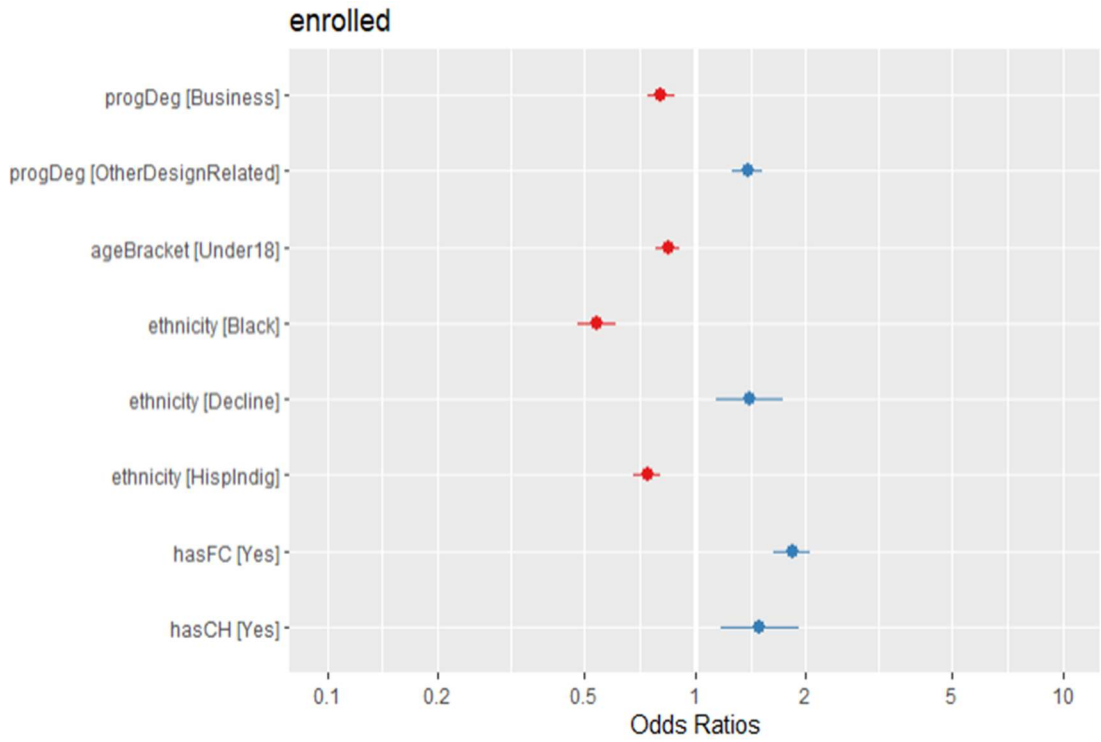

Figure 28. Model B3 exponentiated coefficients

Variable	Odds	2.5%	97.5%
(Intercept)	0.80	0.75	0.86
ProgDeg [Business]	0.81	0.74	0.88

progDeg [OtherDesignRelated]	1.39	1.26	1.53
ageBracket [Under18]	0.84	0.78	0.91
ethnicity [Black]	0.54	0.48	0.61
ethnicity [Decline]	1.40	1.14	1.74
ethnicity [HispIndig]	0.74	0.68	0.80
hasFC [Yes]	1.83	1.63	2.06
hasCH [Yes]	1.49	1.17	1.91

Table 30. Model B3 exponentiated coefficients

The test data was employed in a confusion matrix:

CUTOFF: 0.50	Actual enrollment	
Predicted enrollment	No	Yes
No	1605	982
Yes	212	289

Table 31. Confusion Matrix for predicting enrollment with Model B3

The accuracy was fair, at 61%, but there was a substantial difference in correctly predicting true positives (those who enroll) (Sensitivity=23%) from true negatives (those who do not enroll) (Specificity=88%).

Results when answering Question C) again utilized the expanded set of variables made available now that potential students had submitted applications.

As before, we decided to use logistic regression due to its ease of interpretation, and our large sample size would allow for it ($N_{enrollee}$=6955, with 4648 progressing to Active status). The modeling strategy was backwards elimination because, for the same reasons as described in the results for Question B). Furthermore, Pell was dropped as a variable for the same reasons described in the results for answering Question B).

Model C1: This was the full model. Due to lack of significance, AgeBracket was combined into two levels: Under18 and 18AndOver and under Ethnicity, IndigAmerican was merged with Hispanic into HispIndig. White was not significant and was merged with Asian into AsianWhite. Neither Black nor HispIndig were significant, so they were merged with the baseline Ethnicity variable level AsianWhite. After these adjustments, the Program variable level Business was insignificant. This was merged with the base level Apparel into ApparelBusiness. At this point,

hasFC was marginally insignificant ($p < 0.0512$) and was removed for Model C2. The AIC for Model C1 was 6663.3.

Model C2: The second model had hasFC removed but the AIC increased to 6665.1.

Considering the increased AIC in Model C2, Model C1 was chosen as the final model.

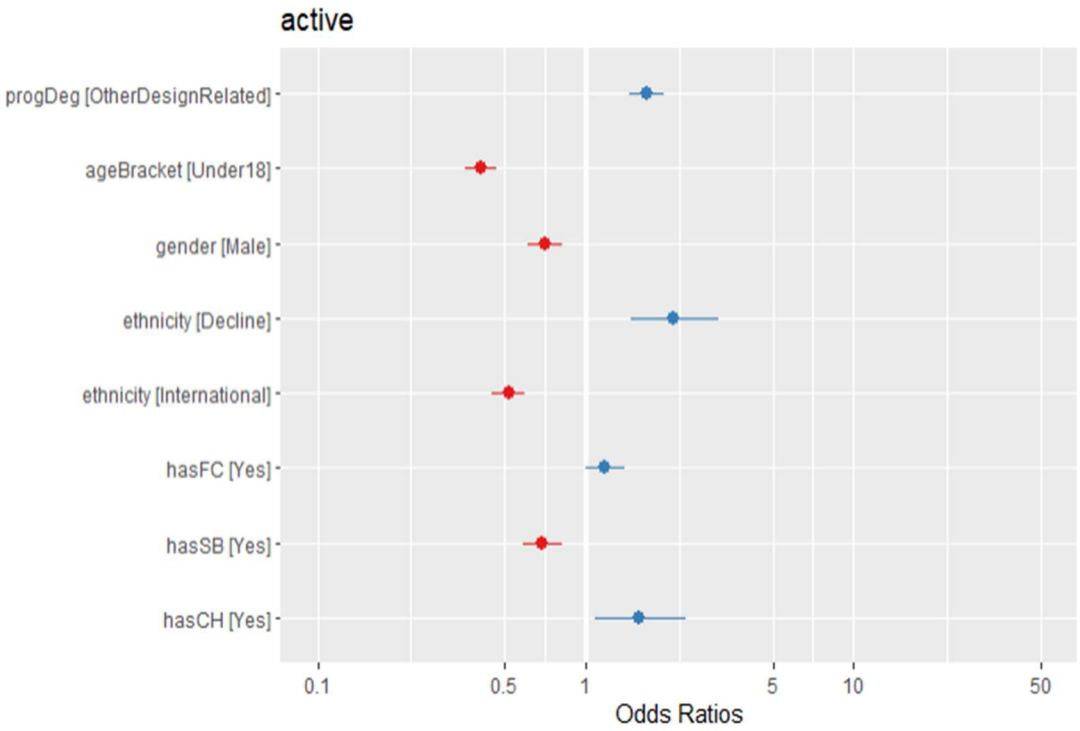

Figure 29. Model C1 exponentiated coefficients

Variable	Odds	2.5%	97.5%
(Intercept)	3.18	2.86	3.54
ProgDeg [OtherDesignRelated]	1.70	1.46	1.97
ageBracket [Under18]	0.41	0.36	0.46
gender [Male]	0.71	0.61	0.82
ethnicity [Decline]	2.13	1.49	3.14
ethnicity [International]	0.52	0.45	0.60
hasFC [Yes]	1.18	1.00	1.40
hasSB [Yes]	0.69	0.58	0.82

hasCH [Yes]	1.59	1.09	2.37

Table 32. Model C1 exponentiated coefficients

The test data was employed in a confusion matrix:

CUTOFF: 0.50	Actual Active status	
Predicted Active status	No	Yes
No	175	112
Yes	286	817

Table 33. Confusion Matrix for predicting Active status with Model C1

The accuracy was fair, at 69%, but there was a difference in correctly predicting true positives (those who progress to Active status) (Sensitivity=95%) from true negatives (those who do not progress to Active status) (Specificity=16%).

Another attempt was made using a different cutoff. There was a noticeable drop in accuracy after a cutoff of about 0.65. In a bid to preserve some accuracy while boosting specificity a cutoff of 0.625 was settled on.

CUTOFF: 0.625	Actual Active status	
Predicted Active status	No	Yes
No	267	317
Yes	194	612

Table 34. Confusion Matrix for predicting Active status with Model C1 (cutoff 0.625)

The accuracy was fair, at 63%, but there was a difference in correctly predicting true positives (those who progress to Active status) (Sensitivity=66%) from true negatives (those who do not progress to Active status) (Specificity=58%).

Results when answering Question D) again utilized the expanded set of variables considering students had submitted applications.

As before, we decided to use logistic regression due to its ease of interpretation, and our large sample size would allow for it (with N_{active}=4554, 1130 withdrew within the first year). The modeling strategy was backwards elimination.

The first model was the full model. Most AgeBracket levels were not significant, so these were releveled to Under18 and 18AndOver. Likewise, almost none of the Ethnicity levels were significant, even after merging the levels Hispanic and Two or more races with IndigAmerican and merging levels Asian and White. A very broad releveling was done to reduce Ethnicity to two levels: one representing traditionally under-represented minorities HispIndig (per Part A. and B.) and Black, and the other representing everyone else. Despite the extensive releveling, neither AgeBracket nor Ethnicity were significant. Also, Pell was insignificant.

The second model removed Ethnicity because it had a greatest p-value. This reduced the AIC from 4125.7 to 4124.3. AgeBracket remained insignificant with a greater p-value than Pell, so it was removed in the third model. This reduced the AIC to 4123.1. Pell remained insignificant, so it was removed in the fourth and final model. The AIC was reduced to 4121.6. The remaining variables Gender and Program were significant.

The coefficients of the resultant model were exponentiated and are shown below:

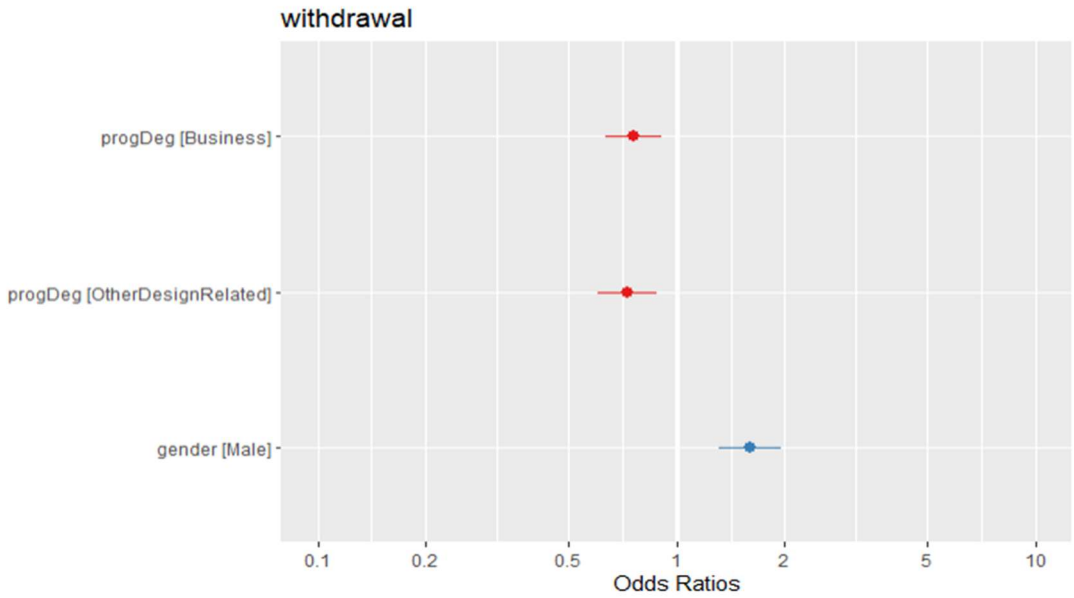

Figure 30. Model D4 exponentiated coefficients

Table 35. Variable	Odds	2.5%	97.5%
(Intercept)	0.38	0.33	0.42
ProgDeg [Business]	0.76	0.63	0.91
progDeg [OtherDesignRelated]	0.73	0.60	0.88
gender [Male]	1.59	1.30	1.94

Table 36. Model D4 exponentiated coefficients

The test data was employed in a confusion matrix:

CUTOFF: 0.50	Actual withdrawal within one year	
Predicted withdrawal within one year	No	Yes
No	676	234
Yes	0	0

Table 37. Confusion Matrix for predicting withdrawal within the first year with Model D4

The accuracy was 75% but the sensitivity was 0%.

A check was done to see if adjusting the cutoff for registering a positive result would improve sensitivity. A cutoff of 0.315 was tried.

CUTOFF: 0.315	Actual withdrawal within one year	
Predicted withdrawal within one year	No	Yes
No	628	203
Yes	48	31

Table 38. Confusion Matrix for predicting withdrawal within the first year with Model D4 (cutoff 0.315)

This lowered the accuracy to 72% but improved the model's detection of true positives. The sensitivity went up to 13% which is still poor.

Additionally, overdispersion was checked, but no indications were found.

CHAPTER 5. DISCUSSION

To consider the results in addressing Question A), most of the top-ranking touchpoints (outside of Online Application where roughly 25% of leads never finished their application) were rare events, even is the ability to drive leads to apply was reasonably good. RFI Phone (WCC) was the highest ranked and the rarest touchpoint (RFI stands for Request for Information). It was difficult to get information on the details of this touchpoint, but indications are it may have changed into something else, possibly Info Call (I), but we chose to keep the two separated because of uncertainty. In any case, the name suggests it was an “incoming” touchpoint. In other words, the lead-initiated contact with the school via this touchpoint. A standing question is: what motivated them? The Event Attendee (EVT) touchpoint existed for a limited number of “one-off” events over the past four years but is not in current use. This would be an example of an “outgoing” touchpoint. The school goes out to the lead or to where they are highly likely to be found. There were about 500 leads with the Junior Advantage LP (JA) touchpoint. This is the landing page on the website for those with the Junior Advantage Merit Scholarship. It’s makes sense this touchpoints success would be reasonably good, considering these are highly motivated future students with a financial advantage. Of the above highlights, a common theme is that the school is not able to affect change with these touchpoints in a substantive manner at this time. This is either because the touchpoints are outdated or the method of gaining applicants from them is passive. What is more interesting are those touchpoints that are currently in play, and either are unduly resource intensive or could have more resources reallocated to them. Of moderate success but greater interest is the Chat touchpoint (CH). Roughly 1000 leads had this touchpoint and it had about a 30% application rate. The Chat touchpoint consists of the chat application on the website. Considering it is not manned 24/7, Marketing has an opportunity to reallocate resources from more expensive touchpoints to improve the Chat’s visibility, responsive time and ease of use. Two other touchpoints of

considerable interest are the Speaker's Bureau touchpoint (SB), and to a lesser extent, the Fashion Club touchpoint (FC). The Speaker's Bureau is a long-running operation and is one of marketing's greatest expenses. Representatives visit high schools across the country for classroom presentations. Over 38 thousand leads have this in their touchpoint trail, which shows the extent of travel that has occurred, but success in those leads subsequently achieving New Application Received status is only about 5% or 1900 leads applying in the 4 years. In comparison, the Fashion Club had nearly 1600 leads applying over 4 years. The greater expense of Speaker's Bureau led to similar results as Fashion Club. Furthermore, Fashion Club is likely to leave a greater impression upon leads because it encourages pro-active involvement from leads in the art and lifestyle of fashion, whereas the Speaker's Bureau does not have the same participatory element.

In resolving what factors were associated with an applicant progressing to New Enrolled / Future Start status (Question B)), the final logistic model did not find odds of enrollment significantly different for the ethnic categories Asian, International, and White. These were grouped together as the baseline level for the Ethnicity variable and relative to this group, the odds of enrolling, given an application was submitted, for Black, Hispanic and indigenous American groups (including Islanders) was significantly lower and their odds were the lowest of any variables in the model, relative to their respective baselines. While we can't know all the reasons potential students fail to progress to the enrollment stage, Student Advising should pay particular attention to this community of applicants. Since the school's business program is competing with many other business programs, it is possible that the lowered odds (19% lower) of progressing to enrollment status compared to the Apparel program could be due to competition from other schools. Looking at the performance of the touchpoints in the model, the Speaker's Bureau (SB) was not significant. This should be of interest to Marketing. Applicants who participated in Fashion Club (FC) at some point were associated with an 83% increased

odds of enrolling. Since the Fashion Club is participatory, this suggests applicants have a high level of interest in the school's programs. Those applicants that had the Chat (CH) touchpoint had a 49% increased likelihood of enrolling. An unknown here is whether the Chat touchpoint was interacted with after applying. If it was, it's possible Chat representatives are guiding applicants through the enrollment process. The final model's performance was good in the area that Student Advising was interested in. Despite how Question B) is posed, and the model is set up, Student Advising wants to know who is dropping out of the Status Pathway for Students at this stage. Using a 0.50 cutoff, the accuracy was modest (61%), but the specificity was very good at 88%. In other words, the model was very good at correctly predicting who would drop-off of the pathway from application to enrollment. Overall, this model could be suitable for predictions in the context of Student Advising focusing their resources on a subset of drop-off risk applicants.

Question C) asks: what factors are associated with an enrolled student progressing to Active status? Despite the fact an applicant reaching enrollment status would suggest a level of stability going forward, as indicated earlier roughly a third of enrollees drop-off before reaching active status. Looking at the final model, both Apparel and Business programs were insignificantly different. Enrollees in OtherDesignRelated programs had 70% increased odds of progressing to Active status, but this was a sort of catch-all category, so it is difficult to comment on. The age bracket Under18 enrollees had the most extreme deviation (59% lower odds), relative to the baseline, of attending their first class. This age cohort tends to be the school's largest, so understanding variety and nature of impediments to this cohort's progress would be worth studying. Males had 29% reduced odds of progressing from enrollment to Active status. While We cannot know all of the reason individuals may drop-off, the male school population is small and it is possible male enrollees may feel they will not be adequately supported in such an environment. None of the Ethnicity levels were significantly different from each other except

Decline (Decline to state) and International. International students who were enrollees had a 48% reduced likelihood of progressing to Active status. One factor at play here is the COVID pandemic. This likely had a significant impact on International students' ability to start their first year of classes. Those enrollees of Ethnicity level Decline showed remarkable increased odd of attending their first class. The odds were over 100% greater. However, nothing else can be said about this group. Similarly, little can be said about the performance of the touchpoints Chat (CH) and Fashion Club (FC) because of their uncertainty. Speaker's Bureau does continue to underperform. This again suggests that the Speaker's Bureau does not generate interest comparable to Fashion Club. As for model performance at a .50 cutoff, the accuracy was modest at 69%, but the ability to correctly detect who would drop-off the pathway from enrollee to in-class student was terrible. This ability came from the specificity which was 16%. The cutoff was adjusted, and a cutoff of .625 gave a compromise between an accuracy and specificity. In this case, accuracy was reduced to 63% and specificity was raised to 58%. This is a modest model, but the seeming zero-sum-game between accuracy and specificity points to research into other variables that could improve measurement as an alternative path.

Finally, Question D): what factors are associated with an active student withdrawing within their first year? This is a highly pertinent question because the withdraw rate over the period of this study was about 25%. The pandemic will have something to do with this number as the school completely shut down all in-person study for a year. A study for another time, but it would be worth looking at the year-by-year withdrawal rate breakdowns. This modeling job highlight how ill-suited many of existing variables were. As it turns out, Age was not significant in whether a student withdrew within the first year. Pell status was not significant, either. Most levels in the Ethnicity variable were insignificant. After merging them into two levels, one representing historically underrepresented minorities (Black, Hispanic, and indigenous Americans (and Islanders)) and the other representing everything else, no significant difference was detected

between the two. Ultimately, Ethnicity was removed. The only variables that remained significant were Program and Gender. Both Business and OtherDesignRelated programs had slightly reduced odds that an active student in either would withdraw within the first year (24% and 27% reduced odds respectively). It was difficult to surmise reasons for this without more knowledge of the individual majors that were grouped under these three broad program classifications. A topic of future study. As for Gender, a male student's odds of withdrawing within the first year were 59% greater than a female student. A possible reason for this was elucidated earlier: because of the majority female population and concomitant infrastructure surrounding them, male students did not feel the same sense of community. For Student Advising, it could be worth exploring increased or alternative male-centric outreach efforts. It could also be worthwhile surveying male student attitudes. Regarding model prediction performance, accuracy with a 50% cutoff was 75%, but with sensitivity 0% and specificity 100%. Student Advising was primarily interested in who will withdraw. As, it stands, this model fails utterly, with 0 correct predictions of withdrawals. Adjusting the cutoff level to .315 improved sensitivity to 18%, but this is still not a reliable model. Aside from offering insight into the performance of male students, attempting the model withdrawals within the first year highlights the need for better measures that can be incorporated into future models.

CHAPTER 6. CONCLUSION

The endeavor of this study was to answer a broad question: What factors relate to a lead passing down the Status Pathway for Students and either persisting as an active student into their second year or withdrawing within the first year? The broad question was split into four stepwise questions, each asking what factors were associated with individuals progressing from one status to the next on the path. At the start of this path, leads first interacted with the Marketing Department through tracked touchpoints that could consist of outreach efforts by the school such as sponsored events or lead-initiated contacts like requests for more information on the school website. The question to answer was: which touchpoints brought leads to application status? This study found that one of the most resource intensive and commonly utilized touchpoints Speaker's Bureau (SB) performed poorly in motivating leads to apply (5% success) when ranked against other similarly resource intensive touchpoints like Fashion Club (FC). Furthermore, the relative success of lower-cost touchpoint Chat (CH) (30% successful driver of applications) and the opportunity to increase its visibility, functionality and responsiveness suggests Marketing should consider shifting resources from Speaker's Bureau to Chat to boost applications coming into the school at less cost. A weakness in this study that could be explored in the future is the lack of data on how touchpoints were found by leads. This is particularly important for touchpoints like Chat that depend on incoming leads. Moving to the next step in the path, demographic data from applicants was used to model who progressed and who dropped-off on the path to enrollee status. Results from the logistic regression model indicated Student Advising should pay close attention to Black, Hispanic and indigenous American groups (including Islanders). Their odds of progressing to enrollment status averaged about 36% lower than other ethnic groups. The model also showed that the touchpoint Fashion Club is bringing in applicants much greater odds of enrolling than Speaker's Club. Student Advising will be interested to know the predictive power of this model was impressive where it counted: correctly

predicting what applicants would drop-off the path. The rate it correctly predicted applicants dropping-off was 88%. Modeling the path from enrollee to active student brough attention to the high drop-off rate for enrollees who were under 18 at the time of application. The odds of becoming an active student decreased by 59% over other age groups. Since the under 18 cohort is the school's largest incoming cohort more resources should be devoted to examining this issue. Student Advising may want to consider expanding outreach efforts with this group of enrollees. Similarly, male students had 29% reduced odds of making it to Active student status. As a distinct minority in the school body, it would be worth exploring whether male enrollees have the perception the school is unable to accommodate their interests. The model did not do a good job predicting who will drop-off the path from enrollment to active student. After loosening the restrictions on who gets counted as dropping off the model correctly predicted drop-offs at a rate of 58%. This was an area where more sophisticated (but less interpretable) modeling techniques should be explored should Student Advising wish for a strong predictive model. The final step in the Status Pathway for students saw a drop-off rate of roughly 25%. It was this percent of active students that withdrew within the first year. The pandemic was a factor and for this reason a more nuanced year-by-year examination of withdrawal rates is recommended. Here, modeling highlighted only the program area students were in and student gender as significant factors in withdrawal rates. Students in the business and other design related majors had reduced odds of withdrawing (24% and 27% respectively) compared to the apparel majors. It is difficult to reach any conclusions here considering the board program categories under examination. Further research should look into the individual majors themselves to find a reason for these drop-offs. Considering male students came up in an earlier model, the fact that they have a 59% greater likelihood of withdrawing within the first year should concern the school, as it suggests that, in fact, male students may indeed feel they are not finding a home on campus. Again, this calls for greater research into male student

perceptions of the school. On the predictive side, this model did a terrible job correctly predicting who would withdraw (a 0% correct prediction of withdrawing rate). However, it performed well correctly predicting who wouldn't withdraw with overall accuracy of 75%. This was not satisfactory, and lowering the threshold for who counted as withdrawing only boosted the rate of correctly predicting withdrawals to 13%. The takeaway here is the nature of data at our disposal was not sufficient to build a robust model of which students withdraw within the first year. One action to take is incorporating first-year academic performance indicators into future models. This study explored the pathway taken by leads as they transition to applicants, enrollees and active students. What could benefit a future broad study of the Status Pathway for Students would be incorporating data on how leads are locating touchpoints. Additionally, incorporating more demographic data such as whether an applicant is the first in their family to attend college or flagging of high schools as a proxy for economic status could improve future models.

APPENDIX A:

Data Set Columns

Phone, Full_Name, and Email were required to join data sets together. Inq.Create_Date and Inquiry_Sequence_Number were used to refine the data set while sub-setting in later steps. AppRecDate, EnrollDate, ActiveDate, and WithdrawalDate were needed to help refine data in later steps and also were used to create the outcome variables. The variable SIS_Collector and variables 11 through 15 were predictor variables. Note that variable names with underscores were sourced from original unaltered spreadsheet. Names with first character capitalized were new variables created in spreadsheet prior to upload into RStudio. Names with camelCase were represent new variables created in RStudio.

#	Variable Name	Type	Description
1	Full_Name	Nominal	Name of lead
2	Phone	Nominal	Phone number of lead
3	Email	Nominal	Email of lead
4	Inq.Create_Date	YYYY-MM-DD	Date of touchpoint (ex. 2021-06-30)
5	Inquiry_Sequence_Number	Ratio (discrete)	The nth touchpoint by a given lead
6	SIS_Collector	Nominal	Code and description of the touchpoint
7	AppRecDate	YYYY-MM-DD	Date an application was received, and the application fee was paid by a lead.
8	EnrollDate	YYYY-MM-DD	Date applicant was enrolled in a program and assigned classes, becoming an enrollee
9	ActiveDate	YYYY-MM-DD	Date enrollee became an active student. This is when they are recorded as attending at least one course
10	WithdrawalDate	YYYY-MM-DD	Date an active student withdraws from school
11	progDeg	Nominal	Program student is pursuing. - Apparel - Business - Other Design Related
12	AgeBracket	Ordinal	Age range of student - Under 18 - 18-19 - 20-21 - 22-24 - 25-29 - 30+
13	Gender	Nominal	Gender of student - Female - Male - Not specified
14	Ethnicity	Nominal	Ethnicity of student / or International if foreign student - American Indian or Alaska Native - Asian - Black or African American

			- Decline to Respond - Hispanic (Hispanic / Latino) - International - Native Hawaiian or Other Pacific Islander - Two or More Races - White
15	Pell	Binary	Did the student receive a Pell grant? - Yes - No

Table A1. Data Set Columns

APPENDIX B

Touchpoint Codes and Definitions

Touchpoint code	Touchpoint description	Touchpoint code	Touchpoint description
WCD2	Career Quiz	JA	Junior Advantage LP
WCQ	Career QuizV2	WMF	Military Friendly LP
WFG	Guess Scholarship	LP	Lecture Partnership
SSC	Live-Study Space Contest	WCB	Chinese Brochure LP
WF	FIDM National Scholarship	WEA	FC Senior LP
LD	Scanned Lecture Leads	WPP	Project Runway LP
LE	Lecture Electronic Card	WPJ	Jacquieline West LP
LEV	Lecture Electronic Card Virtual	LX	Reps at Event
WAI	Admissions Interview	WL1	Fashion Design
WAS	Admissions Saturday Workshop	WL2	Interior Design
WAE	Events	WL3	Merchandise & Marketing
EVT	Event Attendee	WL4	Jewelry Design
WHO	Open House	WL5	Digital Media & Cinema
ASMBR	Alumni Sponsored FC Member	WL6	Visual Communication
ASPRZ	Alumni Sponsored FC President	WL7	Graphic & Visual Design
WESY	FC Registration	WL8	Apparel & Textile Design
WEC	Fashion Club Contests	WL9	Beauty Degrees
WEL	Fashion Club Leader Quiz	WLM	Merchandise Product Development
WESX	FC Application Submit	WLT	Textile Design
FCMBR	FC Member Membership	WLF	Footwear Design & Development
FCPRZ	FC President Membership	WLC	Creative Industry Studies
CCPRZ	FC President Membership College	WLB	Business & MBA
CCMBR	FC Member Membership College	LV	Rep Visitation
WESYC	FC Registration College	WC	Info Request
WESXC	FC Application Submit College	LXV	Rep at Event Virtual
I	Info Call	LC	Collaborative Parnter
X	Walk In	LVV	Rep Visitation Virtual
R	Referral	LTR	Letters
WAC	Click to Chat	TER	Transfer Evaluation Request
L	Lecture Card	WSL	Digital Marketing
N	Non-Lecture Lead	WCE	Mailing RFI
WBL	Online App New Lead	FDB	FIDM Blog
WB	Online App Existing Lead	WCC	RFI Phone
WSF	General Inquiry	WCMD	Info Request Mode Magazine
WTR	Transfer Questions	LI	Lecture International Agency
WC11	Majors LP	CCD	Creative Careers Download
WML	College Rep iPad	TIR	Transfer Information Request

Table B1. Touchpoint codes and definitions

Program	Major_Degree	
Apparel	APPMGMT_AA	FTWRDSGN_AA
	APPMGMT_AdvPD	FTWRDSGN_AdvPD
	APTLDSGN_Bach	IMPD_AdvPD
	BSNDNM_AdvPD	MNSWR_AdvPD
	DESIGN_Bach	MRPRD_AA
	FLMTV_AdvPD	MRPRD_AdvPD
	FSHNDSGN_AA	THTRCSTM_AdvPD
	FSHNDSGN_AdvPD	TXTLDSGN_AA
	FSHNKNT_AA	TXTLDSGN_AdvPD
	FSHNDSGN_AA	
Business	BSNMGMT_Bach	DIGMRK_AdvPD
	BTYMPDR_AA	DIGMRK_Bach
	BTYMPDR_AdvPD	MBA
	BTYBSMG_AA	MRMK_AA
	BTYBSMG_Bach	MRMK_AdvPD
	BTYMRMK_AA	SCLMD_AA
	BTYMRMK_AdvPD	SCLMD_AdvPD
	BTYMGMT_AdvPD	SCLMD_Bach
	DIGMRK_AA	
OtherDesignRelated	CIS_Bach	INTDSGN_AdvPD
	DGTLCNM_Bach	INTDSGN_Bach
	DGTLMD_AA	JWLYDSGN_AA
	DGTLMD_AdvPD	JWLYDSGN_AdvPD
	ENTSETDS_AdvPD	PRFST1_Bach
	GRPHDSGN_AA	PRFST2_Bach
	GRPHDSGN_AdvPD	PRFST3_Bach
	GRPHDSGN_Bach	VSLCMCN_AA
	INTDSGN_AA	VSLCMCN_AdvPD

Table B2. Major and degree under each program level

APPENDIX C

R Code

RStudio 2022.07.1+554
"Spotted Wakerobin" Release (7872775ebddc40635780ca1ed238934c3345c5de, 2022-07-22)
for Windows
Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko)
QtWebEngine/5.12.8 Chrome/69.0.3497.128 Safari/537.36

R version 4.1.3 (2022-03-10) "One Push-Up"

```
######### START LIKELIHOOD TOUCHPOINT ASSOCIATED WITH LEAD APPLYING #########

library("caret")
library("dplyr")
library("forcats")
library("ggplot2")
library("lubridate")
library("readxl")

leads_raw <- read_xlsx("leads_Jun_2017-Oct_1_2022_UPLOAD_nolegacy.xlsx")

# base dataframe of lead data
leads <- leads_raw %>%
  arrange(Email, FIDM_Inquiry_Sequence_Number) %>%
  dplyr::select(Email,
         Inq.Create_Date,
         FIDM_Inquiry_Sequence_Number,
         FIDM_SIS_Collector,
         AppRecDate,
         EnrollDate,
         ActiveDate,
         WithdrawalDate,
         Program,
         Degree,
         AgeBracket,
         Gender,
         Ethnicity,
         Pell,
         ) %>%
  rename(email = Email,
         inquiryDate = Inq.Create_Date,
         inquirySequenceNumber = FIDM_Inquiry_Sequence_Number,
         sisCollector = FIDM_SIS_Collector,
         appRecDate = AppRecDate,
         enrollDate = EnrollDate,
         activeDate = ActiveDate,
         withdrawalDate = WithdrawalDate,
         program = Program,
         degree=Degree,
         ageBracket = AgeBracket,
         gender = Gender,
         ethnicity = Ethnicity,
         pell = Pell,
         ) %>%
  mutate(ethnicity = dplyr::recode(ethnicity,"American Indian or Alaska
Native"="AmericanIndianAlaskaNative","Black or African American"="Black","Decline to
```

```
Respond"="Decline","Native Hawaiian or Other Pacific
Islander"="NativeHawaiianPacificIslander","Two or more races"="TwoOrMoreRaces"),
         gender = dplyr::recode(gender,"Not specified" = "NotSpecified"))

# pull touchpoint code from sisCollector field
if(!'sisCollectorCode' %in% names(leads))
  leads <- leads %>% mutate(sisCollectorCode = gsub( " .*$", "", sisCollector )) %>%
       select(-sisCollector)

leads <- leads %>% mutate(sisCollectorCode = case_when(sisCollectorCode   %in%
c("WB","WBL") ~ "OA", TRUE ~ sisCollectorCode ))

# Group touchpoints and programs
leads <- leads %>%
  mutate(sisCollectorCode = case_when(sisCollectorCode %in% c("WCD2","WCQ") ~ "CQ",
                           sisCollectorCode %in%
c("ASMBR","ASPRZ","WEC","WEL","WESX","FCMBR","FCPRZ","CCPRZ","CCMBR","WESY","WESYC",
"WESXC","WEA") ~ "FC",
                           sisCollectorCode %in%
c("WML","L","LE","LEV","LD","LP","LX","LV","LXV","LVV") ~ "SB",
                           sisCollectorCode %in%
c("WC11","WL1","WL2","WL3","WL5","WL6","WL7","WL8","WL9","WLM","WLC","WLB") ~ "LPM",
# including WC11 as landing page
                           sisCollectorCode %in% c("WAC","WAX") ~ "CH",
                           TRUE ~ sisCollectorCode
                           )
         ) %>%
  mutate(prog_deg = paste0(program,"_",degree)) %>%
  mutate(prog_deg = case_when(prog_deg %in% c("APPMGMT_AA",
                                              "APPMGMT_AdvPD",
                                              "APTLDSGN_Bach",
                                              "BSNDNM_AdvPD",
                                              "DESIGN_Bach",
                                              "FLMTV_AdvPD",
                                              "FSHNDSGN_AA",
                                              "FSHNDSGN_AdvPD",
                                              "FSHNKNT_AA",
                                              "FSHNDSGN_AA",
                                              "FTWRDSGN_AA",
                                              "FTWRDSGN_AdvPD",
                                              "IMPD_AdvPD",
                                              "MNSWR_AdvPD",
                                              "MRPRD_AA",
                                              "MRPRD_AdvPD",
                                              "THTRCSTM_AdvPD",
                                              "TXTLDSGN_AA",
                                              "TXTLDSGN_AdvPD"
                                              ) ~ "Apparel",
                              prog_deg %in% c("CIS_Bach",
                                              "DGTLCNM_Bach",
                                              "DGTLMD_AA",
                                              "DGTLMD_AdvPD",
                                              "ENTSETDS_AdvPD",
                                              "GRPHDSGN_AA",
                                              "GRPHDSGN_AdvPD",
                                              "GRPHDSGN_Bach",
                                              "INTDSGN_AA",
                                              "INTDSGN_AdvPD",
                                              "INTDSGN_Bach",
                                              "JWLYDSGN_AA",
                                              "JWLYDSGN_AdvPD",
                                              "PRFST1_Bach",
```

```
                                                "PRFST2_Bach",
                                                "PRFST3_Bach",
                                                "VSLCMCN_AA",
                                                "VSLCMCN_AdvPD") ~
"OtherDesignRelated",
                              prog_deg %in% c("BSNMGMT_Bach",
                                              "BTYMPDR_AA",
                                              "BTYMPDR_AdvPD",
                                              "BTYBSMG_AA",
                                              "BTYBSMG_Bach",
                                              "BTYMRMK_AA",
                                              "BTYMRMK_AdvPD",
                                              "BTYMGMT_AdvPD",
                                              "DIGMRK_AA",
                                              "DIGMRK_AdvPD",
                                              "DIGMRK_Bach",
                                              "MBA_MBA",
                                              "MRMK_AA",
                                              "MRMK_AdvPD",
                                              "SCLMD_AA",
                                              "SCLMD_AdvPD",
                                              "SCLMD_Bach") ~ "Business",
                             TRUE ~ prog_deg
                             )
         )

# Remove retired touchpoints
leads <- leads %>%filter(!(sisCollectorCode %in% c("CCD","LI", "SSC", "WF", "WL4",
"WLF", "WLT", "WPJ", "WPP")))

# Application to enrollment lag time dataframe
num_applied <- leads %>%
  distinct(email, sisCollectorCode, .keep_all = TRUE) %>%
  #filter(!(sisCollectorCode %in% c("LI", "SSC", "WF", "WL4", "WLF", "WLT", "WPJ",
"WPP"))) %>%
  select(email,inquirySequenceNumber,inquiryDate,sisCollectorCode, appRecDate) %>%
  filter(!is.na(appRecDate) & inquirySequenceNumber == 1 & inquiryDate < appRecDate)
%>%
  group_by(email) %>%
   mutate(
    time_to_apply =
as.numeric(difftime(ymd(appRecDate),ymd(inquiryDate),units="days"))
  )

# Application to enrollment lag time barplot
ggplot(num_applied, aes(x = time_to_apply)) +
  geom_histogram(colour = "black", fill = "white",
                 binwidth = 90) +
  xlab("Days") + ylab("Number of leads")


# Subset using cutoff data for new leads
leads_subset_applied <- leads %>%
  mutate(before_cutoff = ifelse(inquiryDate < "2022-10-01","Yes","No"),
         before_applied = ifelse(is.na(appRecDate) | inquiryDate <=
appRecDate,"Yes","No"),
         applied = ifelse(!is.na(appRecDate),1,0)
         )

# add touchpoint trail column
leads_subset_applied <- leads_subset_applied %>%
  group_by(email) %>%
```

```
  mutate(cutoff_trail = list(unique(before_cutoff,fromLast=TRUE))) %>%
  mutate(before_cutoff = ifelse("Yes" %in% cutoff_trail,"Yes","No")) %>%
  select(-cutoff_trail) %>%
  filter(before_cutoff == "Yes",
    before_applied=="Yes"
    ) %>%
  select(-before_cutoff,
         -before_applied
         )

# remove duplicate lead entries
leads_subset_applied <- distinct(leads_subset_applied, email, sisCollectorCode,
.keep_all = TRUE)

#define one-hot encoding function
dummy <- dummyVars(" ~ sisCollectorCode", data=leads_subset_applied)

#perform one-hot encoding on data frame
leads_subset_applied_tp <- data.frame(leads_subset_applied,predict(dummy,
newdata=leads_subset_applied))

app_grph <- leads_subset_applied_tp %>%
  group_by(email) %>%
  mutate(sisCollectorCodeCH_sum = sum(sisCollectorCodeCH),
         sisCollectorCodeCQ_sum = sum(sisCollectorCodeCQ),
         sisCollectorCodeEVT_sum = sum(sisCollectorCodeEVT),
         sisCollectorCodeFC_sum = sum(sisCollectorCodeFC),
         sisCollectorCodeFDB_sum = sum(sisCollectorCodeFDB),
         sisCollectorCodeI_sum = sum(sisCollectorCodeI),
         sisCollectorCodeJA_sum = sum(sisCollectorCodeJA),
         sisCollectorCodeLC_sum = sum(sisCollectorCodeLC),
         sisCollectorCodeLPM_sum = sum(sisCollectorCodeLPM),
         sisCollectorCodeLTR_sum = sum(sisCollectorCodeLTR),
         sisCollectorCodeN_sum = sum(sisCollectorCodeN),
         sisCollectorCodeOA_sum = sum(sisCollectorCodeOA),
         sisCollectorCodeP_sum = sum(sisCollectorCodeP),
         sisCollectorCodeR_sum = sum(sisCollectorCodeR),
         sisCollectorCodeSB_sum = sum(sisCollectorCodeSB),
         sisCollectorCodeTER_sum = sum(sisCollectorCodeTER),
         sisCollectorCodeTIR_sum = sum(sisCollectorCodeTIR),
         sisCollectorCodeWAE_sum = sum(sisCollectorCodeWAE),
         sisCollectorCodeWC_sum = sum(sisCollectorCodeWC),
         sisCollectorCodeWCC_sum = sum(sisCollectorCodeWCC),
         sisCollectorCodeWCE_sum = sum(sisCollectorCodeWCE),
         sisCollectorCodeWCMD_sum = sum(sisCollectorCodeWCMD),
         sisCollectorCodeWFG_sum = sum(sisCollectorCodeWFG),
         sisCollectorCodeWMF_sum = sum(sisCollectorCodeWMF),
         sisCollectorCodeWSF_sum = sum(sisCollectorCodeWSF),
         sisCollectorCodeWSL_sum = sum(sisCollectorCodeWSL),
         sisCollectorCodeWTR_sum = sum(sisCollectorCodeWTR),
         sisCollectorCodeX_sum = sum(sisCollectorCodeX)
         ) %>%
  distinct(email, .keep_all = TRUE) %>%
  select(-inquirySequenceNumber,
         -sisCollectorCode,
         -sisCollectorCodeCH,
         -sisCollectorCodeCQ,
         -sisCollectorCodeEVT,
         -sisCollectorCodeFC,
         -sisCollectorCodeFDB,
         -sisCollectorCodeI,
         -sisCollectorCodeJA,
```

```
          -sisCollectorCodeLC,
          -sisCollectorCodeLPM,
          -sisCollectorCodeLTR,
          -sisCollectorCodeN,
          -sisCollectorCodeOA,
          -sisCollectorCodeP,
          -sisCollectorCodeR,
          -sisCollectorCodeSB,
          -sisCollectorCodeTER,
          -sisCollectorCodeTIR,
          -sisCollectorCodeWAE,
          -sisCollectorCodeWC,
          -sisCollectorCodeWCC,
          -sisCollectorCodeWCE,
          -sisCollectorCodeWCMD,
          -sisCollectorCodeWFG,
          -sisCollectorCodeWMF,
          -sisCollectorCodeWSF,
          -sisCollectorCodeWSL,
          -sisCollectorCodeWTR,
          -sisCollectorCodeX
        )

code_applied <- c(
           "CH",
          "CQ",
          "EVT",
          "FC",
          "FDB",
          "I",
          "JA",
          "LC",
          "LPM",
          "LTR",
          "N",
          "OA",
          "P",
          "R",
          "SB",
          "TER",
          "TIR",
          "WAE",
          "WC",
          "WCC",
          "WCE",
          'WCMD',
          "WFG",
          "WMF",
          "WSF",
          "WSL",
          "WTR",
          "X"
        )

n_applied_CH    <- sum(app_grph$sisCollectorCodeCH_sum==1)
n_applied_CQ    <- sum(app_grph$sisCollectorCodeCQ_sum==1)
n_applied_EVT   <- sum(app_grph$sisCollectorCodeEVT_sum==1)
n_applied_FC    <- sum(app_grph$sisCollectorCodeFC_sum==1)
n_applied_FDB   <- sum(app_grph$sisCollectorCodeFDB_sum==1)
n_applied_I     <- sum(app_grph$sisCollectorCodeI_sum==1)
n_applied_JA    <- sum(app_grph$sisCollectorCodeJA_sum==1)
n_applied_LC    <- sum(app_grph$sisCollectorCodeLC_sum==1)
```

```
n_applied_LPM   <- sum(app_grph$sisCollectorCodeLPM_sum==1)
n_applied_LTR   <- sum(app_grph$sisCollectorCodeLTR_sum==1)
n_applied_N     <- sum(app_grph$sisCollectorCodeN_sum==1)
n_applied_OA    <- sum(app_grph$sisCollectorCodeOA_sum==1)
n_applied_P     <- sum(app_grph$sisCollectorCodeP_sum==1)
n_applied_R     <- sum(app_grph$sisCollectorCodeR_sum==1)
n_applied_SB    <- sum(app_grph$sisCollectorCodeSB_sum==1)
n_applied_TER   <- sum(app_grph$sisCollectorCodeTER_sum==1)
n_applied_TIR   <- sum(app_grph$sisCollectorCodeTIR_sum==1)
n_applied_WAE   <- sum(app_grph$sisCollectorCodeWAE_sum==1)
n_applied_WC    <- sum(app_grph$sisCollectorCodeWC_sum==1)
n_applied_WCC   <- sum(app_grph$sisCollectorCodeWCC_sum==1)
n_applied_WCE   <- sum(app_grph$sisCollectorCodeWCE_sum==1)
n_applied_WCMD  <- sum(app_grph$sisCollectorCodeWCMD_sum==1)
n_applied_WFG   <- sum(app_grph$sisCollectorCodeWFG_sum==1)
n_applied_WMF   <- sum(app_grph$sisCollectorCodeWMF_sum==1)
n_applied_WSF   <- sum(app_grph$sisCollectorCodeWSF_sum==1)
n_applied_WSL   <- sum(app_grph$sisCollectorCodeWSL_sum==1)
n_applied_WTR   <- sum(app_grph$sisCollectorCodeWTR_sum==1)
n_applied_X     <- sum(app_grph$sisCollectorCodeX_sum==1)

n_applied <- c(
  n_applied_CH,
  n_applied_CQ,
  n_applied_EVT,
  n_applied_FC,
  n_applied_FDB,
  n_applied_I,
  n_applied_JA,
  n_applied_LC,
  n_applied_LPM,
  n_applied_LTR,
  n_applied_N,
  n_applied_OA,
  n_applied_P,
  n_applied_R,
  n_applied_SB,
  n_applied_TER,
  n_applied_TIR,
  n_applied_WAE,
  n_applied_WC,
  n_applied_WCC,
  n_applied_WCE,
  n_applied_WCMD,
  n_applied_WFG,
  n_applied_WMF,
  n_applied_WSF,
  n_applied_WSL,
  n_applied_WTR,
  n_applied_X
)

yes_applied_CH    <- sum(app_grph$sisCollectorCodeCH_sum == 1 & app_grph$applied==1)
yes_applied_CQ    <- sum(app_grph$sisCollectorCodeCQ_sum == 1 & app_grph$applied==1)
yes_applied_EVT   <- sum(app_grph$sisCollectorCodeEVT_sum==1 & app_grph$applied==1)
yes_applied_FC    <- sum(app_grph$sisCollectorCodeFC_sum==1 & app_grph$applied==1)
yes_applied_FDB   <- sum(app_grph$sisCollectorCodeFDB_sum==1 & app_grph$applied==1)
yes_applied_I     <- sum(app_grph$sisCollectorCodeI_sum==1 & app_grph$applied==1)
yes_applied_JA    <- sum(app_grph$sisCollectorCodeJA_sum==1 & app_grph$applied==1)
yes_applied_LC    <- sum(app_grph$sisCollectorCodeLC_sum==1 & app_grph$applied==1)
yes_applied_LPM   <- sum(app_grph$sisCollectorCodeLPM_sum==1 & app_grph$applied==1)
yes_applied_LTR   <- sum(app_grph$sisCollectorCodeLTR_sum==1 & app_grph$applied==1)
```

```
yes_applied_N     <- sum(app_grph$sisCollectorCodeN_sum==1 & app_grph$applied==1)
yes_applied_OA    <- sum(app_grph$sisCollectorCodeOA_sum==1 & app_grph$applied==1)
yes_applied_P     <- sum(app_grph$sisCollectorCodeP_sum==1 & app_grph$applied==1)
yes_applied_R     <- sum(app_grph$sisCollectorCodeR_sum==1 & app_grph$applied==1)
yes_applied_SB    <- sum(app_grph$sisCollectorCodeSB_sum==1 & app_grph$applied==1)
yes_applied_TER   <- sum(app_grph$sisCollectorCodeTER_sum==1 & app_grph$applied==1)
yes_applied_TIR   <- sum(app_grph$sisCollectorCodeTIR_sum==1 & app_grph$applied==1)
yes_applied_WAE   <- sum(app_grph$sisCollectorCodeWAE_sum==1 & app_grph$applied==1)
yes_applied_WC    <- sum(app_grph$sisCollectorCodeWC_sum==1 & app_grph$applied==1)
yes_applied_WCC   <- sum(app_grph$sisCollectorCodeWCC_sum==1 & app_grph$applied==1)
yes_applied_WCE   <- sum(app_grph$sisCollectorCodeWCE_sum==1 & app_grph$applied==1)
yes_applied_WCMD  <- sum(app_grph$sisCollectorCodeWCMD_sum==1 & app_grph$applied==1)
yes_applied_WFG   <- sum(app_grph$sisCollectorCodeWFG_sum==1 & app_grph$applied==1)
yes_applied_WMF   <- sum(app_grph$sisCollectorCodeWMF_sum==1 & app_grph$applied==1)
yes_applied_WSF   <- sum(app_grph$sisCollectorCodeWSF_sum==1 & app_grph$applied==1)
yes_applied_WSL   <- sum(app_grph$sisCollectorCodeWSL_sum==1 & app_grph$applied==1)
yes_applied_WTR   <- sum(app_grph$sisCollectorCodeWTR_sum==1 & app_grph$applied==1)
yes_applied_X     <- sum(app_grph$sisCollectorCodeX_sum==1 & app_grph$applied==1)


yes_applied <- c(
  yes_applied_CH,
  yes_applied_CQ,
  yes_applied_EVT,
  yes_applied_FC,
  yes_applied_FDB,
  yes_applied_I,
  yes_applied_JA,
  yes_applied_LC,
  yes_applied_LPM,
  yes_applied_LTR,
  yes_applied_N,
  yes_applied_OA,
  yes_applied_P,
  yes_applied_R,
  yes_applied_SB,
  yes_applied_TER,
  yes_applied_TIR,
  yes_applied_WAE,
  yes_applied_WC,
  yes_applied_WCC,
  yes_applied_WCE,
  yes_applied_WCMD,
  yes_applied_WFG,
  yes_applied_WMF,
  yes_applied_WSF,
  yes_applied_WSL,
  yes_applied_WTR,
  yes_applied_X
)

p_applied_CH    <- sum(app_grph$sisCollectorCodeCH_sum == 1 &
app_grph$applied==1)/n_applied_CH
p_applied_CQ    <- sum(app_grph$sisCollectorCodeCQ_sum == 1 &
app_grph$applied==1)/n_applied_CQ
p_applied_EVT   <- sum(app_grph$sisCollectorCodeEVT_sum==1 &
app_grph$applied==1)/n_applied_EVT
p_applied_FC    <- sum(app_grph$sisCollectorCodeFC_sum==1 &
app_grph$applied==1)/n_applied_FC
p_applied_FDB   <- sum(app_grph$sisCollectorCodeFDB_sum==1 &
app_grph$applied==1)/n_applied_FDB
```

```
p_applied_I     <- sum(app_grph$sisCollectorCodeI_sum==1 &
app_grph$applied==1)/n_applied_I
p_applied_JA    <- sum(app_grph$sisCollectorCodeJA_sum==1 &
app_grph$applied==1)/n_applied_JA
p_applied_LC    <- sum(app_grph$sisCollectorCodeLC_sum==1 &
app_grph$applied==1)/n_applied_LC
p_applied_LPM   <- sum(app_grph$sisCollectorCodeLPM_sum==1 &
app_grph$applied==1)/n_applied_LPM
p_applied_LTR   <- sum(app_grph$sisCollectorCodeLTR_sum==1 &
app_grph$applied==1)/n_applied_LTR
p_applied_N     <- sum(app_grph$sisCollectorCodeN_sum==1 &
app_grph$applied==1)/n_applied_N
p_applied_OA    <- sum(app_grph$sisCollectorCodeOA_sum==1 &
app_grph$applied==1)/n_applied_OA
p_applied_P     <- sum(app_grph$sisCollectorCodeP_sum==1 &
app_grph$applied==1)/n_applied_P
p_applied_R     <- sum(app_grph$sisCollectorCodeR_sum==1 &
app_grph$applied==1)/n_applied_R
p_applied_SB    <- sum(app_grph$sisCollectorCodeSB_sum==1 &
app_grph$applied==1)/n_applied_SB
p_applied_TER   <- sum(app_grph$sisCollectorCodeTER_sum==1 &
app_grph$applied==1)/n_applied_TER
p_applied_TIR   <- sum(app_grph$sisCollectorCodeTIR_sum==1 &
app_grph$applied==1)/n_applied_TIR
p_applied_WAE   <- sum(app_grph$sisCollectorCodeWAE_sum==1 &
app_grph$applied==1)/n_applied_WAE
p_applied_WC    <- sum(app_grph$sisCollectorCodeWC_sum==1 &
app_grph$applied==1)/n_applied_WC
p_applied_WCC   <- sum(app_grph$sisCollectorCodeWCC_sum==1 &
app_grph$applied==1)/n_applied_WCC
p_applied_WCE   <- sum(app_grph$sisCollectorCodeWCE_sum==1 &
app_grph$applied==1)/n_applied_WCE
p_applied_WCMD  <- sum(app_grph$sisCollectorCodeWCMD_sum==1 &
app_grph$applied==1)/n_applied_WCMD
p_applied_WFG   <- sum(app_grph$sisCollectorCodeWFG_sum==1 &
app_grph$applied==1)/n_applied_WFG
p_applied_WMF   <- sum(app_grph$sisCollectorCodeWMF_sum==1 &
app_grph$applied==1)/n_applied_WMF
p_applied_WSF   <- sum(app_grph$sisCollectorCodeWSF_sum==1 &
app_grph$applied==1)/n_applied_WSF
p_applied_WSL   <- sum(app_grph$sisCollectorCodeWSL_sum==1 &
app_grph$applied==1)/n_applied_WSL
p_applied_WTR   <- sum(app_grph$sisCollectorCodeWTR_sum==1 &
app_grph$applied==1)/n_applied_WTR
p_applied_X     <- sum(app_grph$sisCollectorCodeX_sum==1 &
app_grph$applied==1)/n_applied_X


p_applied <- c(
    p_applied_CH,
  p_applied_CQ,
  p_applied_EVT,
  p_applied_FC,
  p_applied_FDB,
  p_applied_I,
  p_applied_JA,
  p_applied_LC,
  p_applied_LPM,
  p_applied_LTR,
  p_applied_N,
  p_applied_OA,
  p_applied_P,
```

```
  p_applied_R,
  p_applied_SB,
  p_applied_TER,
  p_applied_TIR,
  p_applied_WAE,
  p_applied_WC,
  p_applied_WCC,
  p_applied_WCE,
  p_applied_WCMD,
  p_applied_WFG,
  p_applied_WMF,
  p_applied_WSF,
  p_applied_WSL,
  p_applied_WTR,
  p_applied_X
)



norm_applied <- c(
  n_applied_CH*p_applied_CH & n_applied_CH*(1-p_applied_CH) >= 5,
  n_applied_CQ*p_applied_CQ & n_applied_CQ*(1-p_applied_CQ) >= 5,
  n_applied_EVT*p_applied_EVT & n_applied_EVT*(1-p_applied_EVT) >= 5,
  n_applied_FC*p_applied_FC & n_applied_FC*(1-p_applied_FC) >= 5,
  n_applied_FDB*p_applied_FDB & n_applied_FDB*(1-p_applied_FDB) >= 5,
  n_applied_I*p_applied_I & n_applied_I*(1-p_applied_I) >= 5,
  n_applied_JA*p_applied_JA & n_applied_JA*(1-p_applied_JA) >= 5,
  n_applied_LC*p_applied_LC & n_applied_LC*(1-p_applied_LC) >= 5,
  n_applied_LPM*p_applied_LPM & n_applied_LPM*(1-p_applied_LPM) >= 5,
  n_applied_LTR*p_applied_LTR & n_applied_LTR*(1-p_applied_LTR) >= 5,
  n_applied_N*p_applied_N & n_applied_N*(1-p_applied_N) >= 5,
  n_applied_OA*p_applied_OA & n_applied_OA*(1-p_applied_OA) >= 5,
  n_applied_P*p_applied_P & n_applied_P*(1-p_applied_P) >= 5,
  n_applied_R*p_applied_R & n_applied_R*(1-p_applied_R) >= 5,
  n_applied_SB*p_applied_SB & n_applied_SB*(1-p_applied_SB) >= 5,
  n_applied_TER*p_applied_TER & n_applied_TER*(1-p_applied_TER) >= 5,
  n_applied_TIR*p_applied_TIR & n_applied_TIR*(1-p_applied_TIR) >= 5,
  n_applied_WAE*p_applied_WAE & n_applied_WAE*(1-p_applied_WAE) >= 5,
  n_applied_WC*p_applied_WC & n_applied_WC*(1-p_applied_WC) >= 5,
  n_applied_WCC*p_applied_WCC & n_applied_WCC*(1-p_applied_WCC) >= 5,
  n_applied_WCE*p_applied_WCE & n_applied_WCE*(1-p_applied_WCE) >= 5,
  n_applied_WCMD*p_applied_WCMD & n_applied_WCMD*(1-p_applied_WCMD) >= 5,
  n_applied_WFG*p_applied_WFG & n_applied_WFG*(1-p_applied_WFG) >= 5,
  n_applied_WMF*p_applied_WMF & n_applied_WMF*(1-p_applied_WMF) >= 5,
  n_applied_WSF*p_applied_WSF & n_applied_WSF*(1-p_applied_WSF) >= 5,
  n_applied_WSL*p_applied_WSL & n_applied_WSL*(1-p_applied_WSL) >= 5,
  n_applied_WTR*p_applied_WTR & n_applied_WTR*(1-p_applied_WTR) >= 5,
  n_applied_X*p_applied_X & n_applied_X*(1-p_applied_X) >= 5
)

#qt(0.975,n_applied_LTR-1)*sqrt(p_applied_LTR*(1-p_applied_LTR)/n_applied_LTR)

error_applied_CH      <- qt(0.975,n_applied_CH-1)*sqrt(p_applied_CH*(1-
p_applied_CH)/n_applied_CH)
error_applied_CQ      <- qt(0.975,n_applied_CQ-1)*sqrt(p_applied_CQ*(1-
p_applied_CQ)/n_applied_CQ)
error_applied_EVT     <- qt(0.975,n_applied_EVT-1)*sqrt(p_applied_EVT*(1-
p_applied_EVT)/n_applied_EVT)
error_applied_FC     <- qt(0.975,n_applied_FC-1)*sqrt(p_applied_FC*(1-
p_applied_FC)/n_applied_FC)
error_applied_FDB     <- qt(0.975,n_applied_FDB-1)*sqrt(p_applied_FDB*(1-
p_applied_FDB)/n_applied_FDB)
```

```
error_applied_I      <- qt(0.975,n_applied_I-1)*sqrt(p_applied_I*(1-
p_applied_I)/n_applied_I)
error_applied_JA     <-  qt(0.975,n_applied_JA-1)*sqrt(p_applied_JA*(1-
p_applied_JA)/n_applied_JA)
error_applied_LC     <- qt(0.975,n_applied_LC-1)*sqrt(p_applied_LC*(1-
p_applied_LC)/n_applied_LC)
error_applied_LPM     <- qt(0.975,n_applied_LPM-1)*sqrt(p_applied_LPM*(1-
p_applied_LPM)/n_applied_LPM)
error_applied_LTR     <- qt(0.975,n_applied_LTR-1)*sqrt(p_applied_LTR*(1-
p_applied_LTR)/n_applied_LTR)
error_applied_N      <- qt(0.975,n_applied_N-1)*sqrt(p_applied_N*(1-
p_applied_N)/n_applied_N)
error_applied_OA     <- qt(0.975,n_applied_OA-1)*sqrt(p_applied_OA*(1-
p_applied_OA)/n_applied_OA)
error_applied_P      <- qt(0.975,n_applied_P-1)*sqrt(p_applied_P*(1-
p_applied_P)/n_applied_P)
error_applied_R      <- qt(0.975,n_applied_R-1)*sqrt(p_applied_R*(1-
p_applied_R)/n_applied_R)
error_applied_SB       <- qt(0.975,n_applied_SB-1)*sqrt(p_applied_SB*(1-
p_applied_SB)/n_applied_SB)
error_applied_TER     <- qt(0.975,n_applied_TER-1)*sqrt(p_applied_TER*(1-
p_applied_TER)/n_applied_TER)
error_applied_TIR     <- qt(0.975,n_applied_TIR-1)*sqrt(p_applied_TIR*(1-
p_applied_TIR)/n_applied_TIR)
error_applied_WAE     <- qt(0.975,n_applied_WAE-1)*sqrt(p_applied_WAE*(1-
p_applied_WAE)/n_applied_WAE)
error_applied_WC     <- qt(0.975,n_applied_WC-1)*sqrt(p_applied_WC*(1-
p_applied_WC)/n_applied_WC)
error_applied_WCC     <- qt(0.975,n_applied_WCC-1)*sqrt(p_applied_WCC*(1-
p_applied_WCC)/n_applied_WCC)
error_applied_WCE     <- qt(0.975,n_applied_WCE-1)*sqrt(p_applied_WCE*(1-
p_applied_WCE)/n_applied_WCE)
error_applied_WCMD     <- qt(0.975,n_applied_WCMD-1)*sqrt(p_applied_WCMD*(1-
p_applied_WCMD)/n_applied_WCMD)
error_applied_WFG     <- qt(0.975,n_applied_WFG-1)*sqrt(p_applied_WFG*(1-
p_applied_WFG)/n_applied_WFG)
error_applied_WMF     <- qt(0.975,n_applied_WMF-1)*sqrt(p_applied_WMF*(1-
p_applied_WMF)/n_applied_WMF)
error_applied_WSF     <- qt(0.975,n_applied_WSF-1)*sqrt(p_applied_WSF*(1-
p_applied_WSF)/n_applied_WSF)
error_applied_WSL     <- qt(0.975,n_applied_WSL-1)*sqrt(p_applied_WSL*(1-
p_applied_WSL)/n_applied_WSL)
error_applied_WTR     <- qt(0.975,n_applied_WTR-1)*sqrt(p_applied_WTR*(1-
p_applied_WTR)/n_applied_WTR)
error_applied_X     <- qt(0.975,n_applied_X-1)*sqrt(p_applied_X*(1-
p_applied_X)/n_applied_X)

lb_applied <- c(
  p_applied_CH - error_applied_CH,
  p_applied_CQ - error_applied_CQ,
  p_applied_EVT - error_applied_EVT,
  p_applied_FC - error_applied_FC,
  p_applied_FDB - error_applied_FDB,
  p_applied_I - error_applied_I,
  p_applied_JA - error_applied_JA,
  p_applied_LC - error_applied_LC,
  p_applied_LPM - error_applied_LPM,
  p_applied_LTR - error_applied_LTR,
  p_applied_N - error_applied_N,
  p_applied_OA - error_applied_OA,
  p_applied_P - error_applied_P,
  p_applied_R - error_applied_R,
```

```
  p_applied_SB - error_applied_SB,
  p_applied_TER - error_applied_TER,
  p_applied_TIR - error_applied_TIR,
  p_applied_WAE - error_applied_WAE,
  p_applied_WC - error_applied_WC,
  p_applied_WCC - error_applied_WCC,
  p_applied_WCE - error_applied_WCE,
  p_applied_WCMD - error_applied_WCMD,
  p_applied_WFG - error_applied_WFG,
  p_applied_WMF - error_applied_WMF,
  p_applied_WSF - error_applied_WSF,
  p_applied_WSL - error_applied_WSL,
  p_applied_WTR - error_applied_WTR,
  p_applied_X - error_applied_X
)

ub_applied <- c(
  p_applied_CH + error_applied_CH,
    p_applied_CQ + error_applied_CQ,
  p_applied_EVT + error_applied_EVT,
  p_applied_FC + error_applied_FC,
  p_applied_FDB + error_applied_FDB,
  p_applied_I + error_applied_I,
  p_applied_JA + error_applied_JA,
  p_applied_LC + error_applied_LC,
  p_applied_LPM + error_applied_LPM,
  p_applied_LTR + error_applied_LTR,
  p_applied_N + error_applied_N,
  p_applied_OA + error_applied_OA,
  p_applied_P + error_applied_P,
  p_applied_R + error_applied_R,
  p_applied_SB + error_applied_SB,
  p_applied_TER + error_applied_TER,
  p_applied_TIR + error_applied_TIR,
  p_applied_WAE + error_applied_WAE,
  p_applied_WC + error_applied_WC,
  p_applied_WCC + error_applied_WCC,
  p_applied_WCE + error_applied_WCE,
  p_applied_WCMD + error_applied_WCMD,
  p_applied_WFG + error_applied_WFG,
  p_applied_WMF + error_applied_WMF,
  p_applied_WSF + error_applied_WSF,
  p_applied_WSL + error_applied_WSL,
  p_applied_WTR + error_applied_WTR,
  p_applied_X + error_applied_X
)

tp_applied <-as.data.frame(cbind(code_applied,
                                 no_applied=n_applied-yes_applied,
                                 yes_applied,
                                 n_applied,
                                 p_applied,
                                 norm_applied,
                                 lb_applied,
                                 ub_applied
                                 )
                           ) %>%
  filter(norm_applied==TRUE,lb_applied>0) %>%
  arrange(desc(p_applied))

ggplot(tp_applied, aes(x=fct_inorder(code_applied), y=p_applied, group=1)) +
    geom_errorbar(width=.1, aes(ymin=lb_applied, ymax=ub_applied)) +
```

```
    geom_point(shape=21, size=3, fill="white") +
    ylim(0,100)

######### END LIKELIHOOD TOUCHPOINT ASSOCIATED WITH LEAD APPLYING #########

######### START NEW APPLICATION RECEIVED TO NEW ENROLLED / FUTURE START #########

library("sjPlot")
library("stringr")

newapps <- leads %>%
  select(email,
         inquiryDate,
         sisCollectorCode,
         appRecDate,
         enrollDate,
         progDeg,
         degree,
         #studentState,
         #studentCountry,
         ageBracket,
         gender,
         ethnicity,
         pell,
         #hasFC,
         #hasSB,
         #hasCH
         ) %>%
  filter(!is.na(appRecDate)) %>%
  mutate(
    daysFromAppToEnr =  ifelse(!is.na(enrollDate),
as.numeric(difftime(ymd(enrollDate),ymd(appRecDate),units="days")), -100  ),
    enrolled = ifelse( !is.na(enrollDate) & (daysFromAppToEnr > -30 ) ,"Yes","No")
    )  %>% filter(!(degree %in% c("0","CONTEDU")),
           ageBracket!="0",
           !(gender %in% c("0","NotSpecified")),
           ethnicity!="0",
           !str_detect(progDeg, "0"))

# determine avg lag time
avg_days_to_enroll <- newapps %>%
  group_by(email) %>%
  select(email,appRecDate,enrollDate,daysFromAppToEnr,ageBracket) %>%
  filter(!is.na(enrollDate) & (daysFromAppToEnr > -30 ) ) %>%
  distinct(email,.keep_all = TRUE)

# split by age
avg_days_to_enroll_modAgeBracket <- avg_days_to_enroll %>%
  mutate(ageBracket = case_when(ageBracket %in% c("18-19","20-21","22-24","25-
29","30+") ~ "18AndOver",
                              TRUE ~ ageBracket)
  ) %>%
  mutate(across(ageBracket, factor, levels=c("Under18","18AndOver")))

# means and medians of each age cat.
a <- avg_days_to_enroll %>% filter(ageBracket == "Under18")

print(paste0("The average number of days to enroll after applying for ",nrow(a)," 
applicants under 18 is ",round(mean(a$daysFromAppToEnr),0)," the median is 
",round(median(a$daysFromAppToEnr),0)))
```

```
b <- avg_days_to_enroll %>% filter(ageBracket %in% c("18-19","20-21","22-24","25-
29","30+"))

print(paste0("The average number of days to enroll after applying for ",nrow(b)," 
applicants 18 and over is ",round(mean(b$daysFromAppToEnr),0)," the median is 
",round(median(b$daysFromAppToEnr),0)))

# plot by age
 ggplot(avg_days_to_enroll_modAgeBracket, aes(x = daysFromAppToEnr)) +
   geom_histogram(colour = "black", fill = "white",
                  binwidth = 90) +
   facet_wrap(~ageBracket) + xlab("Days") + ylab("Number of enrollees")

# subset based on lag time
newappsSubset <- newapps %>%
  filter(appRecDate<"2022-10-01")
print(paste0("Removing applicants after 0ct 1, 2022 reduces sample from 
",nrow(newapps)," to ",nrow(newappsSubset)))
print(paste0("Total that enrolled: ",sum(newapps$enrolled=="Yes"), " including 
dups"))

# remove all touchpoints after app. date
newappsSubset <- newappsSubset %>%
  filter(is.na(enrollDate) |
           (!is.na(enrollDate) & inquiryDate <= enrollDate)
  )

# add columns detecting CH, FC and SB in lead touchpoints trail
newappsSubset <- newappsSubset %>%
  group_by(email) %>%
  mutate(leadTrailAppliedPrint = toString(unique(sisCollectorCode))) %>%
  mutate(hasFC = ifelse(grepl("FC",leadTrailAppliedPrint),"Yes","No"),
         hasSB = ifelse(grepl("SB",leadTrailAppliedPrint),"Yes","No"),
         hasCH = ifelse(grepl("CH",leadTrailAppliedPrint),"Yes","No")
  )

# remove dups
newappsSubsetNoDup <- newappsSubset %>%
  distinct(email,.keep_all = TRUE)


###### FILTERING VARIABLE LEVELS FOR INITIAL EDA (COMMENT OUT FOR MODELING) #######
newappsSubsetNoDup <- newappsSubsetNoDup %>%
 mutate(ethnicity = case_when(ethnicity %in% c("Hispanic/Latino","TwoOrMoreRaces") ~ 
"Hispanic",
    ethnicity %in% c("AmericanIndianAlaskaNative","NativeHawaiianPacificIslander") ~ 
"IndigAmerican", TRUE ~ ethnicity
                           )
)


###### FILTERING VARIABLE LEVELS FOR FINAL MODELING (COMMENT OUT FOR EDA) #######
newappsSubsetNoDup <- newappsSubsetNoDup %>%
  mutate(ageBracket = case_when(ageBracket %in% c("18-19","20-21","22-24","25-
29","30+") ~ "18AndOver",TRUE ~ ageBracket),
  ethnicity = case_when(ethnicity %in% 
c("AmericanIndianAlaskaNative","NativeHawaiianPacificIslander","TwoOrMoreRaces","His
panic/Latino") ~ "HispIndig",
                        ethnicity %in% c("Asian","International","White") ~ 
"AsianIntWhite",TRUE ~ ethnicity
                            )
)
```

```
# prep dataframe for modeling
newappsSubsetFactor <- newappsSubsetNoDup %>%
  select(-inquiryDate,-sisCollectorCode,-appRecDate,-enrollDate,-daysFromAppToEnr,-
leadTrailAppliedPrint,-degree)
colnames(newappsSubsetFactor)

newappsSubsetFactor$progDeg <- as.factor(newappsSubsetFactor$progDeg)
newappsSubsetFactor$ageBracket <- as.factor(newappsSubsetFactor$ageBracket)
newappsSubsetFactor$gender <- as.factor(newappsSubsetFactor$gender)
newappsSubsetFactor$ethnicity <- as.factor(newappsSubsetFactor$ethnicity)
newappsSubsetFactor$pell <- as.factor(newappsSubsetFactor$pell)
newappsSubsetFactor$hasFC <- as.factor(newappsSubsetFactor$hasFC)
newappsSubsetFactor$hasSB <- as.factor(newappsSubsetFactor$hasSB)
newappsSubsetFactor$hasCH <- as.factor(newappsSubsetFactor$hasCH)
newappsSubsetFactor$enrolled <- as.factor(newappsSubsetFactor$enrolled)

# 80/20 split for train/test
set.seed(5)
split <- createDataPartition(newappsSubsetFactor$enrolled,p=0.8,list=FALSE)
train_app <- newappsSubsetFactor[split,2:ncol(newappsSubsetFactor)] # skip email
test_app <- newappsSubsetFactor[-split,2:ncol(newappsSubsetFactor)]


# EDA graphs
ggplot(train_app, aes(x = progDeg, fill=enrolled)) +
    geom_bar() +
  ggtitle("Enrollments by program")+ xlab("Program") + ylab("Number of applicants")
+
scale_fill_manual(values=c("#E69F00", "#56B4E9"))

table(train_app$enrolled,train_app$progDeg)

ggplot(train_app, aes(x = fct_relevel(ageBracket,"Under18","18-19","20-21","20-
24","25-29","30+"), fill=enrolled)) +
    geom_bar() +
  ggtitle("Enrollments by Age bracket")+ xlab("Age bracket") + ylab("Number of
applicants") +
  scale_fill_manual(values=c("#E69F00", "#56B4E9"))

table(train_app$enrolled,train_app$ageBracket)

ggplot(train_app, aes(x = gender, fill=enrolled)) +
    geom_bar() +  ggtitle("Enrollments by Gender")+ xlab("Gender") + ylab("Number of
applicants") +
    scale_fill_manual(values=c("#E69F00", "#56B4E9"))

table(train_app$enrolled,train_app$gender)

ggplot(train_app, aes(x = ethnicity, fill=enrolled)) +
    geom_bar() +  ggtitle("Enrollments by Ethnicity")+ xlab("Ethnicity") +
ylab("Number of applicants") +
    scale_fill_manual(values=c("#E69F00", "#56B4E9"))

table(train_app$enrolled,train_app$ethnicity)

ggplot(train_app, aes(x = hasCH, fill=enrolled)) +
    geom_bar() +  ggtitle("Enrollments by Chat touchpoint (CH) status")+ xlab("Has
CH touchpoint") + ylab("Number of applicants") +
    scale_fill_manual(values=c("#E69F00", "#56B4E9"))

table(train_app$enrolled,train_app$hasCH)
```

```
ggplot(train_app, aes(x = hasFC, fill=enrolled)) +
    geom_bar() +  ggtitle("Enrollments by Fashion Club touchpoint (FC) status")+
xlab("Has FC touchpoint") + ylab("Number of applicants") +
    scale_fill_manual(values=c("#E69F00", "#56B4E9"))

table(train_app$enrolled,train_app$hasFC)

ggplot(train_app, aes(x = hasSB, fill=enrolled)) +
    geom_bar() +  ggtitle("Enrollments by Speaker's Bureau touchpoint (SB) status")+
xlab("Has SB touchpoint") + ylab("Number of applicants") +
    scale_fill_manual(values=c("#E69F00", "#56B4E9"))

table(train_app$enrolled,train_app$hasSB)

# model B1 (pell not used)
aem1p <- glm(enrolled~.-pell, data=train_app, family=binomial(link="logit"))
summary(aem1p)

# model B2
aem2p <- glm(enrolled~.-pell-hasSB, data=train_app, family=binomial(link="logit"))
summary(aem2p)

# model B3
aem3p <- glm(enrolled~.-pell-hasSB-gender, data=train_app,
family=binomial(link="logit"))
summary(aem3p)

# B3 exponentiated with margins of error (lvl of sig = 0.05)
round(exp(cbind(Estimate=coef(aem3p),confint(aem3p))),2)

anova(aem3p,aem2p,aem1p,test="Chisq")

plot_model(aem3p)

predicted <- predict(aem3p, test_app, type="response")

confusionMatrix(as.factor(ifelse(predicted>0.5,"Yes","No")),as.factor(test_app$enrol
led),positive="Yes")

######### END NEW APPLICATION RECEIVED TO NEW ENROLLED / FUTURE START #########


######### START NEW ENROLLED / FUTURE START TO ACTIVE #########

newenrs <- leads %>%
  select(email,
         inquiryDate,
         sisCollectorCode,
         enrollDate,
         activeDate,
         progDeg,
         degree,
         #studentState,
         #studentCountry,
         ageBracket,
         gender,
         ethnicity,
         pell,
         #hasFC,
         #hasSB,
         #hasCH
```

```
        ) %>%
  filter(!is.na(enrollDate)) %>%
  mutate(
    daysFromEnrToAct =  ifelse(!is.na(activeDate),
as.numeric(difftime(ymd(activeDate),ymd(enrollDate),units="days")), -100  ),
    active = ifelse( !is.na(activeDate) & (daysFromEnrToAct > -30 ) ,"Yes","No")
    )  %>% filter(!(degree %in% c("0","CONTEDU")),
           ageBracket!="0",
           !(gender %in% c("0","NotSpecified")),
           ethnicity!="0",
           !str_detect(progDeg, "0"))


avg_days_to_active <- newenrs %>%
  group_by(email) %>%
  select(email,enrollDate,activeDate,daysFromEnrToAct,ageBracket) %>%
  filter(!is.na(activeDate) & (daysFromEnrToAct > -30 ) ) %>%
  distinct(email,.keep_all = TRUE)

avg_days_to_active_modAgeBracket <- avg_days_to_active %>%
  mutate(ageBracket = case_when(ageBracket %in% c("18-19","20-21","22-24","25-
29","30+") ~ "18AndOver",
                            TRUE ~ ageBracket)
  ) %>%
  mutate(across(ageBracket, factor, levels=c("Under18","18AndOver")))

# lag time by age bracket plot
 ggplot(avg_days_to_active_modAgeBracket, aes(x = daysFromEnrToAct)) +
   geom_histogram(colour = "black", fill = "white",
                  binwidth = 90) +
   facet_wrap(~ageBracket) + xlab("Days") + ylab("Number active")

# mean/median lag times
a <- avg_days_to_active_modAgeBracket %>% filter(ageBracket == "Under18")

print(paste0("The average number of days to go active after enrollment for
",nrow(a)," enrollees under 18 is ",round(mean(a$daysFromEnrToAct),0)," the median
is ",round(median(a$daysFromEnrToAct),0)))

b <- avg_days_to_active_modAgeBracket %>% filter(ageBracket == "18AndOver")

print(paste0("The average number of days to go active after enrollment for
",nrow(b)," enrollees 18 and over is ",round(mean(b$daysFromEnrToAct),0)," the
median is ",round(median(b$daysFromEnrToAct),0)))

# subset using lag times
newenrsSubset <- newenrs %>%
  filter(ifelse(ageBracket=="Under18",enrollDate<"2022-06-01",enrollDate<"2022-10-
01"))

# remove all touchpoints after active date
newenrsSubset <- newenrsSubset %>%
  filter(is.na(activeDate) |
           (!is.na(activeDate) & inquiryDate <= activeDate)
  )

# add columns detecting CH, FC and SB in lead touchpoints trail
newenrsSubset <- newenrsSubset %>%
  group_by(email) %>%
  mutate(leadTrailAppliedPrint = toString(unique(sisCollectorCode))) %>%
  mutate(hasFC = ifelse(grepl("FC",leadTrailAppliedPrint),"Yes","No"),
         hasSB = ifelse(grepl("SB",leadTrailAppliedPrint),"Yes","No"),
```

```
        hasCH = ifelse(grepl("CH",leadTrailAppliedPrint),"Yes","No")
  )

# remove dups
newenrsSubsetNoDup <- newenrsSubset %>%
  distinct(email,.keep_all = TRUE)

###### FOR EDA ########
newenrsSubsetNoDup <- newenrsSubsetNoDup %>%
 mutate(ethnicity = case_when(ethnicity %in% c("Hispanic/Latino","TwoOrMoreRaces") ~
"Hispanic",
    ethnicity %in% c("AmericanIndianAlaskaNative","NativeHawaiianPacificIslander") ~
"IndigAmerican",
                              TRUE ~ ethnicity
                          )
)




##### FOR MODELING #######
 newenrsSubsetNoDup <- newenrsSubsetNoDup %>%
  mutate(ageBracket = case_when(ageBracket %in% c("18-19","20-21","22-24","25-
29","30+") ~ "18AndOver",
                            TRUE ~ ageBracket),
  ethnicity = case_when(ethnicity %in%
c("AmericanIndianAlaskaNative","NativeHawaiianPacificIslander","TwoOrMoreRaces","His
panic/Latino") ~ "HispIndig",
                          ethnicity %in% c("Asian","White") ~ "AsianWhite",
                                TRUE ~ ethnicity
                           )
 )

 newenrsSubsetNoDup <- newenrsSubsetNoDup %>%
  mutate(ethnicity = case_when(ethnicity %in% c("AsianWhite","Black","HispIndig") ~
"AsianBlackHispIndigWhite",
                              TRUE ~ ethnicity
                           )
 )


 newenrsSubsetNoDup <- newenrsSubsetNoDup %>%
  mutate(progDeg = case_when(progDeg %in% c("Apparel","Business") ~
"ApparelBusiness",
                              TRUE ~ progDeg
                           )
 )

# modeling prep
newenrsSubsetFactor <- newenrsSubsetNoDup %>%
  select(-inquiryDate,-sisCollectorCode,-enrollDate, -activeDate,-daysFromEnrToAct,-
leadTrailAppliedPrint,-degree)

newenrsSubsetFactor$progDeg <- as.factor(newenrsSubsetFactor$progDeg)
newenrsSubsetFactor$ageBracket <- as.factor(newenrsSubsetFactor$ageBracket)
newenrsSubsetFactor$gender <- as.factor(newenrsSubsetFactor$gender)
newenrsSubsetFactor$ethnicity <- as.factor(newenrsSubsetFactor$ethnicity)
newenrsSubsetFactor$pell <- as.factor(newenrsSubsetFactor$pell)
newenrsSubsetFactor$hasFC <- as.factor(newenrsSubsetFactor$hasFC)
newenrsSubsetFactor$hasSB <- as.factor(newenrsSubsetFactor$hasSB)
newenrsSubsetFactor$hasCH <- as.factor(newenrsSubsetFactor$hasCH)
newenrsSubsetFactor$active <- as.factor(newenrsSubsetFactor$active)
```

```
# 80/20 split for train/test
set.seed(5)
split <- createDataPartition(newenrsSubsetFactor$active,p=0.8,list=FALSE)
train_enr <- newenrsSubsetFactor[split,2:ncol(newenrsSubsetFactor)] # skip email
test_enr <- newenrsSubsetFactor[-split,2:ncol(newenrsSubsetFactor)]

# EDA plots
ggplot(train_enr, aes(x = progDeg, fill=active)) +
    geom_bar() +
  ggtitle("Progression to 'active' status by program")+ xlab("Programs") +
ylab("Number of enrollees") +
scale_fill_manual(values=c("#E69F00", "#56B4E9"))

table(train_enr$active,train_enr$progDeg)


ggplot(train_enr, aes(x = fct_relevel(ageBracket,"Under18","18AndOver"),
fill=active)) +
    geom_bar() +
  ggtitle("Progression to 'active' status by Age bracket")+ xlab("Age bracket") +
ylab("Number of enrollees") +
  scale_fill_manual(values=c("#E69F00", "#56B4E9"))

table(train_enr$active,train_enr$ageBracket)

ggplot(train_enr, aes(x = gender, fill=active)) +
    geom_bar() +  ggtitle("Progression to 'active' status by Gender")+
xlab("Gender") + ylab("Number of enrollees") +
    scale_fill_manual(values=c("#E69F00", "#56B4E9"))

table(train_enr$active,train_enr$ethnicity)

ggplot(train_enr, aes(x = ethnicity, fill=active)) +
    geom_bar() +  ggtitle("Progression to 'active' status by Ethnicity")+
xlab("Ethnicity") + ylab("Number of enrollees") +
    scale_fill_manual(values=c("#E69F00", "#56B4E9"))

table(train_enr$active,train_enr$pell)

ggplot(train_enr, aes(x = pell, fill=active)) +
    geom_bar() +  ggtitle("Progression to 'active' status by Pell grant
elegibility")+ xlab("Has Pell eligibility") + ylab("Number of enrollees") +
    scale_fill_manual(values=c("#E69F00", "#56B4E9"))

table(train_enr$active,train_enr$hasCH)

ggplot(train_enr, aes(x = hasCH, fill=active)) +
    geom_bar() +  ggtitle("Progression to 'active' status by Chat touchpoint (CH)
status")+ xlab("Has CH touchpoint") + ylab("Number of enrollees") +
    scale_fill_manual(values=c("#E69F00", "#56B4E9"))

table(train_enr$active,train_enr$hasFC)

ggplot(train_enr, aes(x = hasFC, fill=active)) +
    geom_bar() +  ggtitle("Progression to 'active' status by Fashion Club touchpoint
(FC) status")+ xlab("Has FC touchpoint") + ylab("Number of enrollees") +
    scale_fill_manual(values=c("#E69F00", "#56B4E9"))

table(train_enr$active,train_enr$hasSB)

ggplot(train_enr, aes(x = hasSB, fill=active)) +
```

```
    geom_bar() +  ggtitle("Progression to 'active' status by Speaker's Bureau
touchpoint (SB) status")+ xlab("Has SB touchoint") + ylab("Number of enrollees") +
    scale_fill_manual(values=c("#E69F00", "#56B4E9"))

# Model C1
eam1p <- glm(active~.-pell, data=train_enr, family=binomial(link="logit"))
summary(eam1p)

# Model C2
eam2p <- glm(active~.-pell-hasFC, data=train_enr, family=binomial(link="logit"))
summary(eam2p)

# exponentiated variables
plot_model(eam1p)
round(exp(cbind(Estimate=coef(eam1p),confint(eam1p))),2)

#use model to predict probability of default
predicted <- predict(eam1p, test_enr, type="response")

confusionMatrix(as.factor(ifelse(predicted>0.5,"Yes","No")),as.factor(test_enr$activ
e),positive="Yes")

# Adjusted cutoff
confusionMatrix(as.factor(ifelse(predicted>0.625,"Yes","No")),as.factor(test_enr$act
ive),positive="Yes")

######### END NEW ENROLLED / FUTURE START TO ACTIVE #########


######### ACTIVE TO WITHDRAWAL #########

newactives <- leads %>%
  select(email,
         activeDate,
         withdrawalDate,
         degree,
         progDeg,
         #studentState,
         #studentCountry,
         ageBracket,
         gender,
         ethnicity,
         pell
         ) %>%
  filter(!is.na(activeDate) & activeDate < "2022-01-01") %>%
  mutate(
    daysFromActiveToWithdr = ifelse( !is.na(withdrawalDate),
as.numeric(difftime(ymd(withdrawalDate),ymd(activeDate),units="days")), 1000),
    withdrawal = ifelse( !is.na(withdrawalDate) & !(daysFromActiveToWithdr > 365 |
daysFromActiveToWithdr < 0) ,"Yes","No")
    )   %>% filter(!(degree %in% c("0","CONTEDU")),
           ageBracket!="0",
           !(gender %in% c("0","NotSpecified")),
           ethnicity!="0",
           !str_detect(progDeg, "0")) %>%  #app date can't be greater than enroll
date more than a month
  distinct(email,.keep_all = TRUE)


#USE FOR FINAL MODEL
newactives <- newactives %>%
   mutate(
```

```
  ageBracket = case_when(ageBracket %in% c("18-19","20-21","22-24","25-29","30+") ~
"18AndOver",
                            TRUE ~ ageBracket),
         ethnicity = case_when(
           #ethnicity == "TwoOrMoreRaces" ~ "Hispanic/Latino",
                                ethnicity %in%
c("AmericanIndianAlaskaNative","Black","Hispanic/Latino","NativeHawaiianPacificIslan
der","TwoOrMoreRaces") ~ "HispIndig",
                               ethnicity %in%
c("Asian","Decline","International","White") ~ "Other",
                               #ethnicity %in% c("Black","HispIndig") ~
"BlackHispIndig",
                                          TRUE ~ ethnicity
                                )
              )

# USE FOR EDA
#newactives <- newactives %>%
# mutate(ethnicity = case_when(ethnicity %in% c("Hispanic/Latino","TwoOrMoreRaces")
~ "Hispanic",
#    ethnicity %in% c("AmericanIndianAlaskaNative","NativeHawaiianPacificIslander")
~ "IndigAmerican",
#                                TRUE ~ ethnicity
#                           )
#)

# prep for modeling
newactivesFactor <- newactives  %>%
  select(-email,-degree,-activeDate,-withdrawalDate,-daysFromActiveToWithdr)

newactivesFactor$progDeg <- as.factor(newactivesFactor$progDeg)
newactivesFactor$ageBracket <- as.factor(newactivesFactor$ageBracket)
newactivesFactor$gender <- as.factor(newactivesFactor$gender)
newactivesFactor$ethnicity <- as.factor(newactivesFactor$ethnicity)
newactivesFactor$pell <- as.factor(newactivesFactor$pell)
newactivesFactor$withdrawal <- as.factor(newactivesFactor$withdrawal)

# 80/20 split for train/test
set.seed(55)
split <- createDataPartition(newactivesFactor$withdrawal,p=0.8,list=FALSE)
train_w <- newactivesFactor[split,1:ncol(newactivesFactor)] # skip email
test_w <- newactivesFactor[-split,1:ncol(newactivesFactor)]


ggplot(train_w, aes(x = progDeg, fill=withdrawal)) +
    geom_bar() +
    #geom_bar(position = "dodge") +
#  geom_bar(colour = 4, fill = "white") +
  ggtitle("Withdrawal within 1 year by program")+ xlab("Program") + ylab("Number of
active students") +
scale_fill_manual(values=c("#E69F00", "#56B4E9"))

table(train_w$withdrawal,train_w$progDeg)

# EDA plots
ggplot(train_w, aes(x = fct_relevel(ageBracket,"Under18","18AndOver"),
fill=withdrawal)) +
    geom_bar() +
  ggtitle("Withdrawal within 1 year by Age bracket")+ xlab("Age bracket") +
ylab("Number of active students") +
  scale_fill_manual(values=c("#E69F00", "#56B4E9"))
```

```
table(train_w$withdrawal,train_w$ageBracket)

ggplot(train_w, aes(x = gender, fill=withdrawal)) +
    geom_bar() +  ggtitle("Withdrawal within 1 year by Gender")+ xlab("Gender") +
ylab("Number of active students") +
    scale_fill_manual(values=c("#E69F00", "#56B4E9"))

table(train_w$withdrawal,train_w$gender)

ggplot(train_w, aes(x = ethnicity, fill=withdrawal)) +
    geom_bar() + xlab("Ethnicity") + ylab("Number of active students") +
    scale_fill_manual(values=c("#E69F00", "#56B4E9"))
#+  ggtitle("Withdrawal within 1 year by Ethnicity")+

table(train_w$withdrawal,train_w$ethnicity)

ggplot(train_w, aes(x = pell, fill=withdrawal)) +
    geom_bar() +  ggtitle("Withdrawal within 1 year by Pell grant elegibility")+
xlab("Has Pell eligibility") + ylab("Number of active students") +
    scale_fill_manual(values=c("#E69F00", "#56B4E9"))

table(train_w$withdrawal,train_w$pell)

# Model D1
awm1 <- glm(withdrawal~., data=train_w, family=binomial(link="logit"))
summary(awm1)

# Model D2
awm2 <- glm(withdrawal~.-ethnicity, data=train_w, family=binomial(link="logit"))
summary(awm2)

# Model D3
awm3 <- glm(withdrawal~.-ethnicity-pell, data=train_w,
family=binomial(link="logit"))
summary(awm3)

# Model D4
awm4 <- glm(withdrawal~.-ethnicity-pell-ageBracket, data=train_w,
family=binomial(link="logit"))
summary(awm4)

anova(awm4,awm3,awm2,awm1,test="Chisq")

# exponentiated variables
round(exp(cbind(Estimate=coef(awm4),confint(awm4))),2)
plot_model(awm4)

#use model to predict probability of default
predicted <- predict(awm4, test_w, type="response")
confusionMatrix(as.factor(ifelse(predicted>0.5,"Yes","No")),as.factor(test_w$withdra
wal),positive="Yes")

# adjusted cutoff to 0.315
confusionMatrix(as.factor(ifelse(predicted>0.315,"Yes","No")),as.factor(test_w$withd
rawal),positive="Yes")

# because of model poor performance, check for overdispersion
awm4_od <- glm(withdrawal~progDeg+gender, data=train_w, family=quasibinomial)
pchisq(summary(awm4_od)$dispersion*awm4$df.residual,awm4$df.residual,lower=F)
```

BIBLIOGRAPHY

Bowne, C. (2023). [Lead tracking data from 06/10/2017-09/30/2022 and student tracking data from 06/10/207-12/31/2022] [Unpublished raw data]. Fashion Institute of Design & Merchandising.

Editor, Coughlin, M. A. (Ed.) (2005). Applications of Intermediate Advanced Statistics in Institutional Research. Assn for Institutional Research.

Grolemund, G., & Wickham, H. (2017). R for Data Science. O'Reilly Media.